Finite-window rigidity in Fibonacci numeration

Haobo Ma[1] and Wenlin Zhang*[2]

[1]AELF PTE LTD., #14-02, Marina Bay Financial Centre Tower 1, 8 Marina Blvd, Singapore 018981, Singapore, auric@aelf.io

[2]National University of Singapore, Singapore, e1327962@u.nus.edu

Abstract

For each $m \geq 1$, let $\mathrm{Fold}_m : \{0,1\}^m \to X_m$ send a binary word to the first m digits of the Zeckendorf expansion of its Fibonacci value. We show that Fold_m is the normal-form map of a finite terminating confluent rewrite system and that the family $(\mathrm{Fold}_m)_{m\geq 1}$ is compatible with restriction in m. For $m \geq 3$ we prove that the associated sliding block code Φ_m has the following exact reconstruction property: every block of $n \geq m$ consecutive output symbols has a unique binary preimage of length $n+m-1$. Equivalently, the block map $B_{m,n} : \{0,1\}^{n+m-1} \to \mathcal{L}_n(Y_m)$ is a bijection and $|\mathcal{L}_n(Y_m)| = 2^{n+m-1}$. Equivalently, for $m \geq 3$ the residue vector modulo F_{m+2} of the overlapping local Fibonacci sums determines the entire binary block uniquely.

This block bijection gives local inverse rules with memory r and anticipation $m-1-r$ for every $0 \leq r \leq m-1$, and the one-sided inverse memory $m-1$ is best possible. It also yields explicit formulas for Bernoulli measures pushed forward from the full two-shift: if $\mu_{m,p} = (\Phi_m)_*\nu_p$, then $\mu_{m,p}([a]) = p^{|\omega(a)|_1}(1-p)^{n+m-1-|\omega(a)|_1}$ for $a \in \mathcal{L}_n(Y_m)$, and $H_{\mu_{m,p}}(Y_0^{n-1}) = (n+m-1)h_2(p)$. For Bernoulli input bits, this arithmetic residue vector has an explicit finite-dimensional law and forms an exact m-step Markov process. We also show that the right Fischer cover of Y_m over the natural alphabet X_m is the explicit suffix graph G_m, which has 2^{m-1} states and 2^m edges.

Keywords. Zeckendorf expansion, Fibonacci numeration, finite-window rigidity, Bernoulli transport, exact decoding, Markov order, finite-state normalization.

2020 Mathematics Subject Classification. Primary 11A63; Secondary 11B39, 68Q45, 37B10.

1 Introduction

Zeckendorf expansion singles out a canonical Fibonacci representation of every positive integer. The problem considered here is a finite-window version of that principle. For each $m \geq 1$ we consider the binary window space

$$\Omega_m := \{0,1\}^m.$$

Let

$$X_m := \{w \in \{0,1\}^m : w \text{ contains no subword } 11\},$$

and define

$$\mathrm{Fold}_m(\omega) := \pi_m(Z(N(\omega))),$$

*Corresponding author.

where $N(\omega)$ is the Fibonacci value of ω, $Z(N(\omega))$ is its Zeckendorf expansion, and π_m records the first m digits. Thus Fold_m replaces an arbitrary binary window by its canonical admissible representative in Fibonacci numeration.

Two questions motivate the paper. The first is local: can Fold_m be described by a finite normalization procedure, and how does it behave when the window length changes? The second is reconstructive: if one slides this normalization along a bi-infinite binary sequence, how much of the original sequence can be recovered from the resulting list of overlapping folded windows? In number-theoretic terms, the second question asks whether the vector of overlapping local Fibonacci weighted sums modulo F_{m+2} determines an arbitrary binary block uniquely, and if so, what the sharp observation length is.

We answer the first question by showing that Fold_m is the normal-form map of a finite terminating confluent rewrite system; see Proposition 3.3 and Corollary 3.4. We also prove that the maps Fold_m are compatible with restriction in the window length. In particular, the finite admissible languages form an inverse system whose limit is the one-sided golden-mean shift; see Theorem 4.3.

To study the second question, for each m we consider the sliding block code

$$\Phi_m : \{0,1\}^{\mathbb{Z}} \to X_m^{\mathbb{Z}}$$

with image Y_m, the language of overlapping normalized windows. The main result is that for every $m \geq 3$ and every $n \geq m$, each admissible block of n consecutive folded symbols has a unique raw lift of length $n+m-1$. Thus the threshold m is exactly the point at which local normalization becomes lossless on finite blocks. This yields the block count

$$|\mathcal{L}_n(Y_m)| = 2^{n+m-1} \qquad (n \geq m),$$

and gives a family of local inverse rules obtained by reading any prescribed coordinate of the unique length-m lift. The same block bijection also describes the Bernoulli measures pushed forward from the full two-shift; see Theorem 7.8, Corollary 7.9, Corollary 7.10, and Theorems 7.14, 7.16, and 7.18. At the one-sided endpoint this recovers an inverse with memory $m-1$, and no smaller one-sided memory suffices; see Corollary 7.15.

Thus the same integer m governs several different aspects of the problem. It is the first scale at which overlapping folded windows determine the underlying raw block, the Markov memory of the push-forward of Bernoulli measures, the minimal synchronizing length of the natural presentation, and the size parameter of the right Fischer cover. The symbolic-dynamical language serves mainly to record these consequences efficiently.

Related work

The arithmetic background begins with Zeckendorf's theorem [**?**]. Finite-state aspects of numeration systems were developed systematically by Frougny and collaborators [**?**, **?**, **?**, **?**], while algorithmic forms of Zeckendorf arithmetic, including carry propagation and parallel normalization, were studied in [**?**]. For broader numeration settings one may compare Ostrowski systems and their automata-theoretic realizations [**?**, **?**, **?**], as well as more general work on non-standard and linear-recurrence numeration systems with finite-memory automaton structure and normalization algorithms [**?**, **?**, **?**, **?**]. Recent work has also connected Zeckendorf representations with sequence-level statistical and dynamical phenomena; see [**?**].

The distributional and combinatorial structure of Zeckendorf decompositions has been investigated by Grabner, Tichy, and collaborators from the viewpoint of digital problems and asymptotic distribution of digit sums in non-standard numeration systems [**?**]; by Miller and collaborators, who

established Gaussian behavior for the number of summands in Zeckendorf representations [**?**]; and from the ergodic-theoretic perspective connecting beta-expansions and Parry conditions to symbolic dynamics of numeration [**?**, **?**].

These strands of literature address different questions: recognition of legal expansions, arithmetic operations in non-standard numeration systems, automaticity, distributional behavior, or digit-sum statistics. The present paper does not study addition, parallel arithmetic, or asymptotic digit statistics. Instead it fixes a window length m, normalizes every window locally in Fibonacci numeration, and asks when the original binary block can be reconstructed exactly from the resulting local Zeckendorf data. Equivalently, the maps Φ_m form a family of finite-window factor maps attached to Fibonacci numeration, and the later finite-state constructions record the corresponding determinization data.

More concretely, the work of Frougny and collaborators shows how finite automata recognize admissible expansions and implement normalization or arithmetic in Fibonacci and related numeration systems. Our question is different. We begin with an arbitrary binary block, apply truncated Zeckendorf normalization on every overlapping window of length m, and ask when those local normal forms still determine the original block exactly. Unlike the above literature, the present paper proves a sharp local congruence rigidity threshold and derives its exact consequences for representation counting and random local residue distributions. We are not aware of an earlier result, even in the Fibonacci setting, that identifies this sharp observation length for the recoverability problem or the corresponding threshold in the natural deterministic presentation.

This threshold is of arithmetic interest because it measures how much information survives finite truncation of Zeckendorf normalization. The answer is not merely qualitative. Once the window length reaches m, every admissible folded block of length $n \geq m$ corresponds to exactly one binary block of length $n+m-1$, so $|\mathcal{L}_n(Y_m)| = 2^{n+m-1}$. The same bijection gives explicit push-forward formulas for Bernoulli measures and an exact m-step Markov law on the folded alphabet.

In purely number-theoretic terms, the main results can be stated as follows. For each binary block $\omega = \omega_1 \cdots \omega_L \in \{0,1\}^L$ with $L = n+m-1$, the t-th local Fibonacci sum is $S_t^{(m)}(\omega) := \sum_{j=0}^{m-1} \omega_{t+j} F_{j+2}$, and its residue modulo F_{m+2} is $R_t^{(m)}(\omega) := S_t^{(m)}(\omega) \bmod F_{m+2}$. The residue vector $\mathcal{R}_{m,n}(\omega) := (R_1^{(m)}(\omega), \ldots, R_n^{(m)}(\omega))$ determines ω uniquely. The symbolic-dynamical language provides a convenient framework for organizing the proofs of this and the following consequences.

Theorem 1.1 (Main results)**.** *For every $m \geq 3$, the following statements hold.*

(i) (Local Fibonacci congruence rigidity.) *For every $n \geq m$ and $L = n+m-1$, the map $\mathcal{R}_{m,n} : \{0,1\}^L \to \{0,1,\ldots,F_{m+2}-1\}^n$ that sends a binary block to its vector of overlapping local Fibonacci sums modulo F_{m+2} is injective. Equivalently, the congruence system*

$$\sum_{j=0}^{m-1} \omega_{t+j} F_{j+2} \equiv r_t \pmod{F_{m+2}} \qquad (1 \leq t \leq n)$$

has at most one solution $\omega \in \{0,1\}^L$ for each $(r_1, \ldots, r_n) \in \{0,1,\ldots,F_{m+2}-1\}^n$.

(ii) (Exact block reconstruction.) *For every $n \geq m$, every admissible block of n consecutive folded symbols has a unique raw lift of length $n+m-1$. Equivalently, the block map $B_{m,n} : \{0,1\}^{n+m-1} \to \mathcal{L}_n(Y_m)$ is a bijection. Consequently $|\mathcal{L}_n(Y_m)| = 2^{n+m-1}$ for every $n \geq m$, and the n-block presentation of the normalized language is canonically the $(n+m-1)$-block presentation of the full binary shift.*

(iii) (Exact bit recovery from residue windows.) *Write $\mathfrak{R}_{m,m}$ for the set of realizable length-m residue vectors. For each $0 \le s \le m-1$, there exists a function $\rho_{m,s} : \mathfrak{R}_{m,m} \to \{0,1\}$ such that*

$$\omega_{t+s} = \rho_{m,s}\big(R_t^{(m)}(\omega), \dots, R_{t+m-1}^{(m)}(\omega)\big)$$

for every $\omega \in \{0,1\}^{n+m-1}$ and every valid index t. The minimum number of consecutive residues required for any one-sided recovery rule is exactly m.

(iv) (Explicit law and exact Markov order.) *For every $p \in (0,1)$, if ν_p denotes the Bernoulli product law on $\{0,1\}^{\mathbb{Z}}$ and $\mu_{m,p} := (\Phi_m)_*\nu_p$, then every cylinder $[a]_0 \subset Y_m$ with $a \in \mathcal{L}_n(Y_m)$ and $n \ge m$ satisfies the explicit formula*

$$\mu_{m,p}([a]_0) = p^{|\omega(a)|_1}(1-p)^{n+m-1-|\omega(a)|_1}, \qquad \omega(a) := B_{m,n}^{-1}(a).$$

Hence $H_{\mu_{m,p}}(Y_0^{n-1}) = (n+m-1)h_2(p)$, the excess entropy equals $(m-1)h_2(p)$, and the coordinate process under $\mu_{m,p}$ is exact m-step Markov. The random local Fibonacci residue process $R_t := (\sum_{j=0}^{m-1} X_{t+j}F_{j+2}) \bmod F_{m+2}$ under i.i.d. Bernoulli(p) input is itself an exact m-step Markov process.

(v) (Sliding block inverses.) *Exact recovery from overlapping normalized windows is possible from any length-m observation window: for each $0 \le r \le m-1$ there is a sliding block inverse with memory r and anticipation $m-1-r$. At the one-sided endpoint $r = m-1$, no smaller memory suffices.*

(vi) (Fischer cover and presentation complexity.) *Over the natural alphabet X_m, the canonical minimal right-resolving presentation of Y_m is the explicit suffix graph G_m, so it has exactly 2^{m-1} states and 2^m edges. Since $|X_m| = F_{m+2}$, the natural deterministic presentation size grows exponentially relative to the alphabet size.*

Taken together, these results show that a single integer controls the finite-window arithmetic of the problem: it is the point at which local Zeckendorf normalization becomes lossless on blocks, the sharp threshold for local Fibonacci congruence rigidity, the exact Markov memory of the random residue process, and the scale parameter of the natural deterministic presentation; see again Corollary 7.28 and Theorem 7.27.

These statements are proved in Theorem 7.8, Corollary 7.9, Corollary 7.10, Theorem 7.14, Theorem 7.16, Corollary 7.17, Theorem 7.18, Corollary 7.15, Theorem 7.13, Corollary 7.7, Proposition 7.24, and Corollary 7.26, together with Theorem 7.27.

Sketch of the reconstruction mechanism. The inverse is built in two steps. First, the pairwise overlap maps Θ_m compress adjacent folded symbols to the fold of their common raw overlap; iterating this reduction through a block $a_1 \cdots a_m \in \mathcal{L}_m(Y_m)$ yields the central bit $\Theta_{m,m}(a_1 \cdots a_m)$. Second, once this central bit and the folded inner overlaps are known, the one-step left- and right-resolving properties of G_m force the neighboring bits uniquely, so an outward induction recovers the entire raw lift of length $2m-1$. The inverse family in Section 7 simply records the same reconstruction after choosing which coordinate of the recovered length-m raw window is read as the output bit.

The paper is organized as follows. Sections 2 and 3 develop finite-window normalization and its rewrite description. Section 4 proves compatibility in the window length, Section 5 formulates the overlap-decoding problem, Section 6 records the explicit finite-state presentation used later, and Section 7 proves the block reconstruction theorem, derives the measure-theoretic consequences, and then studies the natural right-resolving presentation of the folded language. Section 8 reformulates

the main results in purely number-theoretic terms: local Fibonacci congruence rigidity, representation fingerprint separation, exact bit recovery from residue windows, and the precise distribution and Markov order of random local residue vectors.

2 Finite-window normalization

We introduce the finite languages and the map Fold_m. The only numeration-theoretic input needed later is Zeckendorf uniqueness, which identifies the golden-mean language as the canonical target of the fold.

2.1 Admissible words

Throughout the paper the Fibonacci numbers are normalized by

$$F_1 = F_2 = 1, \qquad F_{k+2} = F_{k+1} + F_k \quad (k \geq 1).$$

For $m \geq 1$ and $s \in \{0, 1\}$ let

$$X_m^{(s)} := \{w = w_1 \cdots w_m \in X_m : w_m = s\}.$$

Thus $X_m^{(0)}$ and $X_m^{(1)}$ split admissible words by their last symbol.

Proposition 2.1 (Terminal recursion for admissible words)**.** *For every $m \geq 1$, if*

$$v_m := \begin{pmatrix} |X_m^{(1)}| \\ |X_m^{(0)}| \end{pmatrix},$$

then

$$v_{m+1} = N_\tau v_m, \qquad N_\tau = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

Proof. If $w \in X_{m+1}$ ends in 1, then its penultimate symbol must be 0. Deleting the last symbol therefore gives a bijection

$$X_{m+1}^{(1)} \longleftrightarrow X_m^{(0)},$$

so $|X_{m+1}^{(1)}| = |X_m^{(0)}|$.

If $w \in X_{m+1}$ ends in 0, then deleting the last symbol gives an arbitrary word in X_m, and appending 0 to a word in X_m never creates the forbidden pattern 11. Hence

$$|X_{m+1}^{(0)}| = |X_m| = |X_m^{(0)}| + |X_m^{(1)}|.$$

Combining the two identities gives the displayed matrix recursion. □

Corollary 2.2 (Fibonacci counting)**.** *For every $m \geq 1$ one has*

$$|X_m| = F_{m+2}.$$

Proof. The recursion in Proposition 2.1 is the Fibonacci recursion with the initial vector

$$v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Therefore

$$|X_m| = |X_m^{(0)}| + |X_m^{(1)}| = F_{m+2}.$$ □

2.2 Fibonacci value and the fold

Let

$$X_\infty^{\text{fin}} := \left\{ c = (c_1, c_2, \ldots) \in \{0,1\}^{\mathbb{N}} : \operatorname{supp}(c) \text{ finite and } c_k c_{k+1} = 0 \text{ for all } k \right\}.$$

For $c \in X_\infty^{\text{fin}}$ define

$$\operatorname{Val}(c) := \sum_{k \geq 1} c_k F_{k+1}.$$

Zeckendorf's theorem gives, for every $N \in \mathbb{N}$, a unique $Z(N) \in X_\infty^{\text{fin}}$ satisfying $\operatorname{Val}(Z(N)) = N$ [**?**].

For a finite binary window $\omega = \omega_1 \cdots \omega_m \in \Omega_m$ define its Fibonacci value by

$$N(\omega) := \sum_{k=1}^{m} \omega_k F_{k+1}.$$

Let

$$\pi_m : X_\infty^{\text{fin}} \to X_m, \qquad \pi_m(c_1, c_2, \ldots) := (c_1, \ldots, c_m),$$

be the length-m prefix projection.

Definition 2.3 (Finite-window fold)**.** For $m \geq 1$ define

$$\operatorname{Fold}_m : \Omega_m \to X_m, \qquad \operatorname{Fold}_m(\omega) := \pi_m(Z(N(\omega))).$$

Definition 2.4 (Visible value)**.** For $x = x_1 \cdots x_m \in X_m$ define

$$V_m(x) := \sum_{k=1}^{m} x_k F_{k+1}.$$

Proposition 2.5 (Visible-value parametrization of X_m)**.** *For every $m \geq 1$, the map*

$$V_m : X_m \to \{0, 1, \ldots, F_{m+2} - 1\}$$

is a bijection.

Proof. If $x \in X_m$, then

$$(x_1, \ldots, x_m, 0, 0, \ldots) \in X_\infty^{\text{fin}}$$

is already a Zeckendorf digit string, so by uniqueness it is the Zeckendorf representation of the integer $V_m(x)$. In particular $V_m(x) < F_{m+2}$, because otherwise the Zeckendorf representation would have to use the weight F_{m+2} or a larger one.

Conversely, let $0 \leq r < F_{m+2}$. The Zeckendorf representation $Z(r)$ cannot use the weight F_{m+2} or any larger weight, so it has the form

$$Z(r) = (x_1, \ldots, x_m, 0, 0, \ldots)$$

for a unique $x \in X_m$. Then $V_m(x) = r$. Thus V_m is bijective. □

Corollary 2.6 (Unique overflow decomposition)**.** *For every $m \geq 1$ and every $\omega \in \Omega_m$, there exists a unique $b(\omega) \in \{0,1\}$ such that*

$$N(\omega) = V_m(\operatorname{Fold}_m(\omega)) + b(\omega) F_{m+2}.$$

Equivalently, $\operatorname{Fold}_m(\omega)$ *is the unique admissible word in X_m whose visible value is congruent to $N(\omega)$ modulo F_{m+2}.*

symbol	domain	role
$N(\omega)$	Ω_m	full Fibonacci value of the raw window ω
$V_m(x)$	X_m	visible value carried by the admissible word x
$b(\omega)$	Ω_m	overflow bit beyond the visible range $[0, F_{m+2})$
$\mathrm{Val}(a)$	$\mathcal{D}$	value of a finitely supported digit configuration

Table 1: Value maps used throughout the paper. The visible value stays within the length-m window, while the overflow bit records whether one copy of F_{m+2} has escaped beyond it.

Proof. Since

$$0 \le N(\omega) \le \sum_{k=1}^{m} F_{k+1} = F_{m+3} - 2 < 2F_{m+2},$$

there is a unique $b(\omega) \in \{0, 1\}$ such that

$$r := N(\omega) - b(\omega)F_{m+2}$$

lies in $\{0, 1, \ldots, F_{m+2} - 1\}$. By Proposition 2.5, there is a unique $x \in X_m$ with $V_m(x) = r$. The first m digits of $Z(N(\omega))$ are precisely this x, so $x = \mathrm{Fold}_m(\omega)$. □

For convenience, Table 1 summarizes the value maps used throughout the paper.

3 A local rewrite description

In this section we describe Fold_m by local rewriting on finitely supported configurations. This gives a concrete normalization procedure before we pass to inverse limits and sequence-level dynamics.

3.1 Value-preserving local rules

Let

$$\mathcal{D} := \{a = (a_1, a_2, \ldots) : a_k \in \mathbb{N},\ \mathrm{supp}(a) \text{ finite}\},$$

and define

$$\mathrm{Val}(a) := \sum_{k \ge 1} a_k F_{k+1}.$$

We write e_k for the kth standard basis vector and

$$\iota_m(\omega) := (\omega_1, \ldots, \omega_m, 0, 0, \ldots) \in \mathcal{D}$$

for the natural embedding of a window into $\mathcal{D}$.

The rewrite system acts on $\mathcal{D}$ through the following local rules:

$$e_k + e_{k+1} \longrightarrow e_{k+2} \qquad (k \ge 1), \tag{3.1}$$

$$2e_1 \longrightarrow e_2, \tag{3.2}$$

$$2e_2 \longrightarrow e_1 + e_3, \tag{3.3}$$

$$2e_k \longrightarrow e_{k-2} + e_{k+1} \qquad (k \ge 3). \tag{3.4}$$

Each rule is local and preserves Val because it is exactly the Fibonacci identity

$$F_{k+1} + F_{k+2} = F_{k+3}, \qquad 2F_2 = F_3, \qquad 2F_3 = F_2 + F_4, \qquad 2F_{k+1} = F_{k-1} + F_{k+2}.$$

Lemma 3.1 (Fibonacci rigidity of value-preserving local weights)**.** *Suppose a positive weight sequence* $(w_k)_{k\geq 1}$ *has the property that each rewrite rule* (3.1)–(3.4) *preserves the weighted sum*

$$\mathrm{Val}_w(a) := \sum_{k\geq 1} a_k w_k \qquad (a \in \mathcal{D}).$$

Then

$$w_{k+2} = w_{k+1} + w_k \qquad (k \geq 1).$$

In particular, under the normalization $w_1 = F_2 = 1$ *and* $w_2 = F_3 = 2$*, one has* $w_k = F_{k+1}$ *for all* $k \geq 1$*.*

Proof. Preservation of (3.1) gives

$$w_k + w_{k+1} = w_{k+2} \qquad (k \geq 1),$$

which is exactly the Fibonacci recursion. The remaining rules are then automatic consequences of that recursion:

$$2w_1 = w_2, \qquad 2w_2 = w_1 + w_3, \qquad 2w_k = w_{k-2} + w_{k+1} \quad (k \geq 3).$$

With the stated normalization, the recursion has the unique solution $w_k = F_{k+1}$. □

Lemma 3.2 (Local weight rigidity)**.** *Let*

$$A(a) := \sum_{k\geq 1} a_k, \qquad B(a) := \sum_{k\geq 1} k a_k, \qquad C(a) := a_2,$$

and order triples lexicographically. Then every rewrite step $a \to a'$ *generated by* (3.1)–(3.4) *strictly decreases*

$$\mu(a) := (A(a), B(a), C(a)).$$

In particular the rewrite system is strongly terminating.

Proof. Rules (3.1) and (3.2) reduce the total digit count A by one. Rule (3.4) for $k \geq 3$ preserves A but changes B by

$$2k - \big((k-2) + (k+1)\big) = 1,$$

so B decreases by one. Rule (3.3) preserves both A and B but decreases $C = a_2$ by two. Hence every rewrite step strictly decreases $\mu(a)$. □

3.2 Canonical normal forms

Proposition 3.3 (Termination, confluence, and normal forms)**.** *The rewrite system on* $\mathcal{D}$ *generated by* (3.1)–(3.4) *is strongly terminating and confluent. Every* $a \in \mathcal{D}$ *therefore has a unique normal form*

$$\mathrm{NF}(a) \in X_\infty^{\mathrm{fin}}$$

satisfying

$$\mathrm{Val}(\mathrm{NF}(a)) = \mathrm{Val}(a).$$

Proof. Strong termination is Lemma 3.2. A configuration is irreducible exactly when every digit is in $\{0,1\}$ and no adjacent pair 11 occurs; this is precisely the definition of X_∞^{fin}.

Fix $a \in \mathcal{D}$. By termination, every maximal reduction sequence from a ends in some irreducible configuration $x \in X_\infty^{\mathrm{fin}}$. Since each rewrite preserves Val, every such terminal configuration satisfies

$$\mathrm{Val}(x) = \mathrm{Val}(a).$$

If x and y are two terminal reductions of a, then $x, y \in X_\infty^{\mathrm{fin}}$ are irreducible configurations with the same Fibonacci value. Zeckendorf uniqueness therefore forces $x = y$. Thus every a has a unique irreducible descendant. For a terminating rewrite system, uniqueness of irreducible descendants is equivalent to confluence, so the system is confluent and the normal form $\mathrm{NF}(a)$ is well defined. □

Corollary 3.4 (Order-independent implementation of the fold)**.** *For every $\omega \in \Omega_m$,*

$$\mathrm{Fold}_m(\omega) = \pi_m(\mathrm{NF}(\iota_m(\omega))),$$

and the right-hand side is independent of the order in which rewrites are applied.

Proof. By Proposition 3.3, the normal form of $\iota_m(\omega)$ is the unique admissible Fibonacci digit string with value $N(\omega)$, namely $Z(N(\omega))$. Taking the first m digits gives the claim. □

Remark 3.5 (Two illustrative normalizations)*.* The following two examples illustrate the order-independence of the rewriting and the appearance of the overflow bit.

For $\omega = 1110 \in \Omega_4$ one may reduce

$$\iota_4(\omega) = e_1 + e_2 + e_3 \to 2e_3 \to e_1 + e_4,$$

so $\mathrm{NF}(\iota_4(\omega)) = \iota_4(1001)$ and

$$\mathrm{Fold}_4(1110) = 1001.$$

If instead $\omega = 1101 \in \Omega_4$, then

$$\iota_4(\omega) = e_1 + e_2 + e_4 \to e_3 + e_4 \to e_5,$$

hence

$$\mathrm{Fold}_4(1101) = 0000, \qquad N(1101) = F_6 = F_{4+2}.$$

This is the smallest example in which the visible word is zero while the whole value survives as the overflow bit from Corollary 2.6.

Proposition 3.6 (Basic properties of Fold_m)**.** *For every $m \geq 1$, the map $\mathrm{Fold}_m : \Omega_m \to X_m$ is well defined and satisfies:*

(i) $\mathrm{Fold}_m(\omega) \in X_m$ *for every* $\omega \in \Omega_m$*;*

(ii) $\mathrm{Fold}_m(x) = x$ *for every* $x \in X_m$*;*

(iii) $\mathrm{Fold}_m : \Omega_m \twoheadrightarrow X_m$ *is surjective.*

Proof. Property (i) is immediate from Definition 2.3. For (ii), if $x \in X_m$ then x is already an admissible Zeckendorf prefix, so $Z(N(x)) = (x_1, \ldots, x_m, 0, 0, \ldots)$ and hence $\mathrm{Fold}_m(x) = x$. Surjectivity follows from (ii): every $x \in X_m$ is a fixed point of Fold_m. □

4 Compatibility in the window length

We next compare the folds at different window lengths. The results of this section show that admissible restriction and finite-window normalization are compatible with one another.

4.1 Restriction maps

Definition 4.1 (Prefix restrictions)**.** For integers $n \geq m \geq 1$ define

$$\pi_{n\to m} : X_n \to X_m, \qquad \pi_{n\to m}(x_1 \cdots x_n) := x_1 \cdots x_m.$$

Lemma 4.2 (Functoriality of admissible restrictions)**.** *If $n \geq \ell \geq m \geq 1$, then*

$$\pi_{n\to m} = \pi_{\ell\to m} \circ \pi_{n\to \ell}.$$

Moreover, for every $c \in X_\infty^{\mathrm{fin}}$ and every $n \geq m$,

$$\pi_{n\to m}(\pi_n(c)) = \pi_m(c).$$

Proof. All maps involved are literal prefix truncations, so both identities are immediate. □

Theorem 4.3 (Inverse-limit identification)**.** *Let*

$$X_\infty := \{c = (c_1, c_2, \ldots) \in \{0,1\}^{\mathbb{N}} : c_k c_{k+1} = 0 \text{ for all } k \geq 1\}.$$

Then there is a natural bijection

$$X_\infty \cong \varprojlim (X_m, \pi_{m+1\to m}).$$

Proof. Given $c \in X_\infty$, its prefix family

$$(\pi_m(c))_{m\geq 1}$$

is compatible under the restriction maps, hence defines an element of the inverse limit. Conversely, given a compatible family

$$(x^{(m)})_{m\geq 1} \in \varprojlim (X_m, \pi_{m+1\to m}),$$

define $c_k := x_k^{(m)}$ for any $m \geq k$. Compatibility ensures that this is independent of the choice of m. Since every finite prefix $x^{(m)}$ avoids 11, the resulting sequence c lies in X_∞. The two constructions are inverse to one another. □

Remark 4.4. The object in Theorem 4.3 is one sided because the resolution tower is indexed by prefix length. It should not be confused with the two-sided image shift Y_m introduced later at the sequence level.

4.2 Compatibility with raw windows

If one truncates a raw window directly, folding does not in general commute with restriction. Let

$$r_{m+1\to m} : \Omega_{m+1} \to \Omega_m, \qquad r_{m+1\to m}(\omega_1 \cdots \omega_{m+1}) = \omega_1 \cdots \omega_m.$$

For $m = 1$ and $\omega = 11$ one has

$$(\mathrm{Fold}_1 \circ r_{2\to 1})(11) = \mathrm{Fold}_1(1) = 1,$$

whereas

$$(\pi_{2\to 1} \circ \mathrm{Fold}_2)(11) = 0,$$

since $N(11) = F_2 + F_3 = F_4$ and hence $\mathrm{Fold}_2(11) = 00$.

The correct raw-level restriction is therefore the folded one.

Definition 4.5 (Fold-compatible raw restriction)**.** For $m \geq 1$ define

$$\widetilde{r}_{m+1\to m} : \Omega_{m+1} \to \Omega_m, \qquad \widetilde{r}_{m+1\to m}(\omega) := \pi_{m+1\to m}(\mathrm{Fold}_{m+1}(\omega)).$$

Proposition 4.6 (Commuting diagram for fold-compatible restriction)**.** *For every $m \geq 1$ and every $\omega \in \Omega_{m+1}$,*

$$\mathrm{Fold}_m(\widetilde{r}_{m+1\to m}(\omega)) = \pi_{m+1\to m}(\mathrm{Fold}_{m+1}(\omega)).$$

Equivalently, the diagram

$$\begin{tikzcd} \Omega_{m+1} \arrow[r, "\mathrm{Fold}_{m+1}"] \arrow[d, "\widetilde{r}_{m+1\to m}"'] & X_{m+1} \arrow[d, "\pi_{m+1\to m}"] \\ \Omega_m \arrow[r, "\mathrm{Fold}_m"] & X_m \end{tikzcd}$$

commutes.

Proof. By definition,

$$\widetilde{r}_{m+1\to m}(\omega) = \pi_{m+1\to m}(\mathrm{Fold}_{m+1}(\omega)) \in X_m.$$

Applying Proposition 3.6(ii) to this admissible word gives

$$\mathrm{Fold}_m(\widetilde{r}_{m+1\to m}(\omega)) = \widetilde{r}_{m+1\to m}(\omega) = \pi_{m+1\to m}(\mathrm{Fold}_{m+1}(\omega)).$$ □

Remark 4.7 (Two raw-level restriction examples)*.* The discrepancy between direct restriction and folded restriction already appears at the smallest scale, where $m = 1$ and $\omega = 11$:

$$(\mathrm{Fold}_1 \circ r_{2\to 1})(11) = 1, \qquad (\mathrm{Fold}_1 \circ \widetilde{r}_{2\to 1})(11) = 0.$$

At the next nontrivial scale one may take $m = 3$ and $\omega = 1101$. Then

$$r_{4\to 3}(1101) = 110, \qquad \mathrm{Fold}_3(110) = 001,$$

whereas Remark 3.5 gives $\mathrm{Fold}_4(1101) = 0000$, so

$$\widetilde{r}_{4\to 3}(1101) = 000, \qquad \mathrm{Fold}_3(\widetilde{r}_{4\to 3}(1101)) = 000.$$

Thus the naive truncation retains a visible carry that disappears once one first folds at the larger resolution and only then restricts.

5 Overlapping normalized windows

We now pass to bi-infinite sequences by sliding the window along the integers. This places the construction in the standard framework of factor maps between shift spaces.

Let σ denote the left shift on both $\{0,1\}^{\mathbb{Z}}$ and $X_m^{\mathbb{Z}}$.

Definition 5.1 (Sequence-level fold)**.** For $m \geq 1$ define

$$\Phi_m : \{0,1\}^{\mathbb{Z}} \to X_m^{\mathbb{Z}}, \qquad (\Phi_m(s))_t := \mathrm{Fold}_m(s_t s_{t+1} \cdots s_{t+m-1}).$$

Proposition 5.2 (Finite radius)**.** *The map Φ_m is a sliding block code with memory* 0 *and anticipation $m-1$.*

Proof. By definition, the symbol $(\Phi_m(s))_t$ depends only on the coordinates $s_t, \ldots, s_{t+m-1}$. □

Proposition 5.3 (Shift equivariance)**.** *For every $m \geq 1$,*

$$\Phi_m \circ \sigma = \sigma \circ \Phi_m.$$

Proof. Both sides evaluate Fold_m on the same translated length-m window at each time t. □

Proposition 5.4 (Factor property on subshifts)**.** *Let $S \subset \{0,1\}^{\mathbb{Z}}$ be a subshift. Then $\Phi_m|_S : S \to X_m^{\mathbb{Z}}$ is continuous, shift commuting, and onto its image*

$$Y_m(S) := \Phi_m(S).$$

Moreover $Y_m(S)$ is a subshift of $X_m^{\mathbb{Z}}$, so

$$\Phi_m|_S : S \twoheadrightarrow Y_m(S)$$

is a factor map.

Proof. Continuity follows from Proposition 5.2, and shift-equivariance from Proposition 5.3. Since S is compact and shift invariant, its image under the continuous equivariant map Φ_m is again compact and shift invariant. By definition, the restriction is onto that image. □

Definition 5.5 (The image shift)**.** We write

$$Y_m := \Phi_m(\{0,1\}^{\mathbb{Z}}) \subset X_m^{\mathbb{Z}}.$$

6 Finite-state structure of normalized windows

This section records the explicit suffix graph attached to the folded language. It will be used throughout the reconstruction arguments in the final section.

6.1 The graph G_m

Definition 6.1 (Suffix-state graph)**.** For $m \geq 1$, let

$$V_m := \{0,1\}^{m-1}.$$

For each $v = v_1 \cdots v_{m-1} \in V_m$ and each $b \in \{0,1\}$ define an edge

$$e = (v, b) : v \to v' := v_2 \cdots v_{m-1} b,$$

labeled by

$$\ell_m(e) := \mathrm{Fold}_m(v_1 \cdots v_{m-1} b) \in X_m.$$

The resulting finite labeled graph is denoted

$$G_m = (V_m, E_m, \ell_m).$$

It has $|V_m| = 2^{m-1}$ vertices and $|E_m| = 2^m$ edges.

Theorem 6.2 (Suffix-graph presentation of the folded language)**.** *Let $Z_m \subset E_m^{\mathbb{Z}}$ be the two-sided edge shift of G_m, and let*

$$\mathcal{L}_m : Z_m \to X_m^{\mathbb{Z}}$$

be the edge-label map. Then

$$Y_m = \mathcal{L}_m(Z_m).$$

In particular G_m presents the folded language Y_m.

Proof. Take any $s \in \{0,1\}^{\mathbb{Z}}$ and define

$$v_t := s_t \cdots s_{t+m-2} \in V_m, \qquad e_t := (v_t, s_{t+m-1}) \in E_m.$$

The compatibility of successive windows implies that $(e_t)_{t\in\mathbb{Z}}$ is a bi-infinite path in G_m, and by construction

$$(\mathcal{L}_m(e))_t = \ell_m(e_t) = \mathrm{Fold}_m(s_t \cdots s_{t+m-1}) = (\Phi_m(s))_t.$$

Hence $Y_m \subset \mathcal{L}_m(Z_m)$.

Conversely, let $e = (e_t)_{t\in\mathbb{Z}} \in Z_m$. Writing each edge as $e_t = (v_t, b_t)$, path compatibility says that v_{t+1} is obtained from v_t by deleting its first symbol and appending b_t. Therefore the family of suffixes v_t and appended bits b_t determines a unique binary sequence $s \in \{0,1\}^{\mathbb{Z}}$ such that

$$s_t \cdots s_{t+m-1} = v_t b_t$$

for all t. Then

$$(\Phi_m(s))_t = \mathrm{Fold}_m(v_t b_t) = \ell_m(e_t) = (\mathcal{L}_m(e))_t.$$

Thus $\mathcal{L}_m(Z_m) \subset Y_m$, proving equality. □

Corollary 6.3 (The image shift is sofic)**.** *For every $m \geq 1$, the image shift Y_m is a sofic shift.*

Proof. This is immediate from Theorem 6.2, which gives an explicit finite labeled graph presenting Y_m. □

6.2 Deterministic structure of the presentation

We begin with the one-step resolving properties of the graph G_m.

Proposition 6.4 (One-step distinguishability)**.** *For every $m \geq 2$ and every $v \in \{0,1\}^{m-1}$,*

$$\mathrm{Fold}_m(v0) \neq \mathrm{Fold}_m(v1).$$

Consequently G_m is right-resolving: the two outgoing edges from any fixed vertex carry distinct labels.

Proof. Let $\omega_0 := v0$ and $\omega_1 := v1$. Then

$$N(\omega_1) - N(\omega_0) = F_{m+1}.$$

If $\mathrm{Fold}_m(\omega_0) = \mathrm{Fold}_m(\omega_1)$, then Corollary 2.6 implies

$$\begin{aligned} N(\omega_1) - N(\omega_0) &= \big(b(\omega_1) - b(\omega_0)\big)F_{m+2} \\ &\in \{-F_{m+2}, 0, F_{m+2}\}, \end{aligned}$$

because the visible value $V_m(\mathrm{Fold}_m(\omega_i))$ would be the same for $i = 0,1$ and only the overflow bit could differ. This contradicts $F_{m+1} > 0$ and $F_{m+1} \neq F_{m+2}$. Hence the two labels are distinct. □

Proposition 6.5 (One-step left-resolving property)**.** *For every $m \geq 2$ and every $v \in \{0,1\}^{m-1}$,*

$$\mathrm{Fold}_m(0v) \neq \mathrm{Fold}_m(1v).$$

Consequently the two incoming edges to any fixed vertex of G_m carry distinct labels.

Proof. Let $\omega_0 := 0v$ and $\omega_1 := 1v$. Then

$$N(\omega_1) - N(\omega_0) = F_2 = 1.$$

If $\mathrm{Fold}_m(\omega_0) = \mathrm{Fold}_m(\omega_1)$, then Corollary 2.6 implies

$$N(\omega_1) - N(\omega_0) = \big(b(\omega_1) - b(\omega_0)\big)F_{m+2} \in \{-F_{m+2}, 0, F_{m+2}\},$$

which is impossible because $F_{m+2} \geq 2$ for $m \geq 2$. Hence the labels are distinct. Applied to the two incoming edges of a fixed vertex $v = v_1 \cdots v_{m-1}$, this says the edges labeled by the windows $0v_1 \cdots v_{m-1}$ and $1v_1 \cdots v_{m-1}$ are distinguished. □

Corollary 6.6 (Entropy of the image shift)**.** *For every $m \geq 2$,*

$$h_{\mathrm{top}}(Y_m) = \log 2.$$

Proof. Since Φ_m is a factor of the full two-shift,

$$h_{\mathrm{top}}(Y_m) \leq \log 2.$$

On the other hand, every vertex of G_m has exactly two outgoing edges, so the number of length-n paths starting from any fixed vertex is 2^n. By Proposition 6.4, two such paths cannot have the same label sequence. Hence Y_m has at least 2^n allowed blocks of length n, so $h_{\mathrm{top}}(Y_m) \geq \log 2$. □

6.3 Fibres and synchronizing blocks

For $m \geq 1$ and $0 \leq r \leq \lfloor m/2 \rfloor$ define

$$z_{m,0} := 0^m, \qquad z_{m,r} := 0^{m-2r}11(01)^{r-1} \quad (1 \leq r \leq \lfloor m/2 \rfloor).$$

Proposition 6.7 (The zero fibre of 0^m)**.** *For every $m \geq 1$,*

$$\mathrm{Fold}_m^{-1}(0^m) = \{z_{m,r} : 0 \leq r \leq \lfloor m/2 \rfloor\}.$$

In particular,

$$\big|\mathrm{Fold}_m^{-1}(0^m)\big| = \lfloor m/2 \rfloor + 1.$$

Proof. By Corollary 2.6,

$$\mathrm{Fold}_m(\omega) = 0^m \quad \Longleftrightarrow \quad N(\omega) \in \{0, F_{m+2}\}.$$

The case $N(\omega) = 0$ forces $\omega = 0^m$, so it remains to classify

$$A_m := \{\omega \in \Omega_m : N(\omega) = F_{m+2}\}.$$

For $m = 1$ the set A_1 is empty, since the only available values are 0 and $F_2 = 1 < F_3$. Fix now $m \geq 2$ and let $\omega \in A_m$. Then

$$\sum_{k=1}^{m-1} \omega_k F_{k+1} \leq \sum_{k=1}^{m-1} F_{k+1} = F_{m+2} - 2 < F_{m+2},$$

so necessarily $\omega_m = 1$. If also $\omega_{m-1} = 1$, then

$$\sum_{k=1}^{m-2} \omega_k F_{k+1} = F_{m+2} - F_{m+1} - F_m = 0,$$

hence $\omega = 0^{m-2}11$. If instead $\omega_{m-1} = 0$, write $\omega = u01$ with $u \in \Omega_{m-2}$. Then

$$N(u) = F_{m+2} - F_{m+1} = F_m,$$

so $u \in A_{m-2}$. Therefore, for every $m \geq 2$,

$$A_m = \{0^{m-2}11\} \cup \{u01 : u \in A_{m-2}\}.$$

Starting from $A_1 = \varnothing$ and $A_2 = \{11\}$ and unwinding this recursion gives

$$A_m = \{z_{m,r} : 1 \leq r \leq \lfloor m/2 \rfloor\}.$$

Adding the zero word yields the stated formula for $\mathrm{Fold}_m^{-1}(0^m)$. The cardinality follows immediately. □

Definition 6.8 (Alternating extremal words)**.** For each $m \geq 1$, let $a_m, b_m \in X_m$ denote the unique alternating admissible words of length m ending respectively in 0 and 1. Thus

$$a_1 = 0, \qquad b_1 = 1,$$

and for $m \geq 2$,

$$a_m = b_{m-1}0, \qquad b_m = a_{m-1}1.$$

Lemma 6.9 (Extremal values of the alternating words)**.** *For every $m \geq 1$ one has*

$$V_m(a_m) = F_{m+1} - 1, \qquad V_m(b_m) = F_{m+2} - 1.$$

Proof. We argue by induction on m. For $m = 1$ the claim is

$$V_1(a_1) = 0 = F_2 - 1, \qquad V_1(b_1) = 1 = F_3 - 1.$$

Assume now that $m \geq 2$ and the claim is known for $m - 1$. Since $a_m = b_{m-1}0$, one has

$$V_m(a_m) = V_{m-1}(b_{m-1}) = F_{m+1} - 1.$$

Likewise, using $b_m = a_{m-1}1$,

$$V_m(b_m) = V_{m-1}(a_{m-1}) + F_{m+1} = (F_m - 1) + F_{m+1} = F_{m+2} - 1.$$

□

Proposition 6.10 (Singleton fibres of the alternating symbols)**.** *For every $m \geq 1$,*

$$\mathrm{Fold}_m^{-1}(a_m) = \{a_m\}, \qquad \mathrm{Fold}_m^{-1}(b_m) = \{b_m\}.$$

Proof. The case $m = 1$ is immediate. Fix $m \geq 2$.

Suppose first that $\mathrm{Fold}_m(\omega) = a_m$. By Lemma 6.9 and Corollary 2.6,

$$N(\omega) \in \{F_{m+1} - 1,\ F_{m+1} + F_{m+2} - 1\}.$$

But

$$F_{m+1} + F_{m+2} - 1 = F_{m+3} - 1 > F_{m+3} - 2 \geq N(\omega),$$

so necessarily $N(\omega) = F_{m+1} - 1$. In particular $N(\omega) < F_{m+1}$, and therefore $\omega_m = 0$. Writing $\omega = u0$ with $u \in \Omega_{m-1}$, we get

$$N(u) = F_{m+1} - 1.$$

By the induction hypothesis, $u = b_{m-1}$. Hence $\omega = b_{m-1}0 = a_m$.

Now suppose that $\mathrm{Fold}_m(\omega) = b_m$. Again by Lemma 6.9 and Corollary 2.6,

$$N(\omega) \in \{F_{m+2} - 1,\ 2F_{m+2} - 1\}.$$

Since

$$2F_{m+2} - 1 > F_{m+2} + F_{m+1} - 2 = F_{m+3} - 2 \geq N(\omega),$$

one must have $N(\omega) = F_{m+2} - 1$. The contribution of the first $m - 1$ digits is at most

$$\sum_{k=1}^{m-1} F_{k+1} = F_{m+2} - 2,$$

so necessarily $\omega_m = 1$. Writing $\omega = u1$ with $u \in \Omega_{m-1}$, we obtain

$$N(u) = F_{m+2} - 1 - F_{m+1} = F_m - 1.$$

By the induction hypothesis, $u = a_{m-1}$. Therefore

$$\omega = a_{m-1}1 = b_m.$$

□

Theorem 6.11 (Single-symbol synchronization)**.** *For every $m \geq 1$, each of the symbols a_m and b_m labels exactly one edge of G_m. Consequently both a_m and b_m are synchronizing words for the presentation G_m of Y_m.*

Proof. An edge of G_m is labeled by $x \in X_m$ precisely when its defining raw word lies in $\mathrm{Fold}_m^{-1}(x)$. Proposition 6.10 shows that $\mathrm{Fold}_m^{-1}(a_m)$ and $\mathrm{Fold}_m^{-1}(b_m)$ are singletons. Hence each of a_m and b_m occurs on exactly one edge of G_m. Any label carried by exactly one edge is synchronizing, since reading that label collapses the candidate vertex set to the terminal vertex of the unique edge. □

Corollary 6.12 (The image language has synchronizing symbols)**.** *For every $m \geq 1$, the symbols a_m and b_m are synchronizing for the presentation G_m. In particular, the image language Y_m is synchronized.*

Proof. By Proposition 6.4, the graph G_m is a right-resolving presentation of Y_m. By Theorem 6.11, that presentation admits a synchronizing symbol. Hence Y_m is synchronized; see [**?**, Ch. 3] or [**?**, Ch. 4]. □

6.4 A lower bound for one-sided inversion

Proposition 6.13 (A lower bound for one-sided inversion)**.** *Let $m \geq 3$, and define finite binary words*

$$u_m := 0^m 1 0^{m-2}, \qquad v_m := 0^{m-2} 1 1 0^{m-1}$$

of length $2m - 1$. Let $U_m, V_m \in \{0,1\}^{\mathbb{Z}}$ be their zero extensions on the coordinates $\{1, \ldots, 2m-1\}$. Also define

$$e_{m,0} := 0^m, \qquad e_{m,j} := 0^{m-j} 1 0^{j-1} \quad (1 \leq j \leq m-1).$$

Then

$$(\Phi_m(U_m))_t = (\Phi_m(V_m))_t = e_{m,t-1} \qquad (1 \le t \le m-1),$$

while

$$(\Phi_m(U_m))_m = 010^{m-2} \neq 10^{m-1} = (\Phi_m(V_m))_m.$$

Consequently, if

$$\Psi : Y_m \to \{0,1\}^{\mathbb{Z}}$$

is a sliding block code with memory q and anticipation 0 such that

$$\Psi \circ \Phi_m = \mathrm{id}_{\{0,1\}^{\mathbb{Z}}},$$

then necessarily $q \ge m-1$.

Proof. For U_m, the tth raw window is $e_{m,t-1} \in X_m$ for every $1 \le t \le m-1$, so

$$(\Phi_m(U_m))_t = e_{m,t-1} \qquad (1 \le t \le m-1).$$

For V_m, the first raw window is $0^{m-2}11 = z_{m,1}$, so Proposition 6.7 gives

$$(\Phi_m(V_m))_1 = 0^m = e_{m,0}.$$

For $2 \le t \le m-1$, the tth raw window of V_m is

$$0^{m-t-1}110^{t-1}.$$

A single application of the local rule (3.1) transforms this word into $e_{m,t-1} = 0^{m-t+1}10^{t-2} \in X_m$, so Corollary 3.4 yields

$$(\Phi_m(V_m))_t = e_{m,t-1} \qquad (2 \le t \le m-1).$$

Thus the first $m-1$ output symbols of $\Phi_m(U_m)$ and $\Phi_m(V_m)$ agree.

At time $t = m$, the two raw windows are

$$010^{m-2} \qquad \text{and} \qquad 10^{m-1}.$$

Both words already lie in X_m, and they are distinct, proving the displayed inequality at time m.

Now suppose that Ψ has memory q and anticipation 0 and satisfies $\Psi \circ \Phi_m = \mathrm{id}$. Set

$$\widetilde{U}_m := \sigma^{m-1}U_m, \qquad \widetilde{V}_m := \sigma^{m-1}V_m.$$

Then

$$(\widetilde{U}_m)_0 = 0, \qquad (\widetilde{V}_m)_0 = 1,$$

because u_m and v_m first differ at their $(m-1)$st coordinate. By Proposition 5.3, the output sequences $\Phi_m(\widetilde{U}_m)$ and $\Phi_m(\widetilde{V}_m)$ agree on the coordinates $-(m-2), \dots, 0$. If $q < m-1$, then the local rule defining $\Psi(\cdot)_0$ reads only a block contained in those common coordinates, so

$$\Psi(\Phi_m(\widetilde{U}_m))_0 = \Psi(\Phi_m(\widetilde{V}_m))_0.$$

Since $\Psi \circ \Phi_m = \mathrm{id}$, this would force $(\widetilde{U}_m)_0 = (\widetilde{V}_m)_0$, a contradiction. Therefore $q \ge m-1$. $\square$

state v	appended bit b	label $\ell_3(v,b)$	next state
00	0	000	00
00	1	001	01
01	0	010	10
01	1	000	11
10	0	100	00
10	1	101	01
11	0	001	10
11	1	100	11

Table 2: The labeled graph G_3.

Proposition 6.14 (Base case at length 3)**.** *Every block*

$$a_1a_2a_3 \in \mathcal{L}_3(Y_3)$$

has a unique raw lift

$$\omega_1 \cdots \omega_5 \in \Omega_5$$

satisfying

$$\mathrm{Fold}_3(\omega_j\omega_{j+1}\omega_{j+2}) = a_j \qquad (1 \le j \le 3).$$

Consequently there is a well-defined map

$$\Theta_{3,3} : \mathcal{L}_3(Y_3) \to X_1 = \{0,1\}, \qquad \Theta_{3,3}(a_1a_2a_3) := \omega_3.$$

Proof. Table 2 gives the complete labeled graph G_3. A block $a_1a_2a_3 \in \mathcal{L}_3(Y_3)$ is carried by some length-3 path in G_3. Since G_3 has four states and two outgoing edges from each state, there are

$$4 \cdot 2^3 = 32$$

length-3 paths in total. A direct inspection of the 32 paths encoded by Table 2 shows that no two distinct length-3 paths carry the same label word. Hence every admissible block $a_1a_2a_3$ determines a unique length-3 path in G_3.

If the initial vertex of that path is $\omega_1\omega_2$ and its successive appended bits are $\omega_3, \omega_4, \omega_5$, then the path corresponds to the unique raw lift $\omega_1 \cdots \omega_5$. The central bit ω_3 is therefore uniquely determined, and this defines $\Theta_{3,3}$. □

7 Finite-window rigidity and consequences

We now prove the main reconstruction theorem. Once one knows that a raw block is determined by overlapping folded windows, the block counts, local inverse rules, push-forward measure formulas, and minimal presentation results follow from the same argument.

7.1 Reconstruction from overlaps

Here $\mathcal{L}_n(Y_m)$ denotes the set of length-n blocks occurring in Y_m.

Lemma 7.1 (A signed Fibonacci gap shifts by one index)**.** *Let $m \ge 2$ and let $d_1, \dots, d_{m-1} \in \{-1, 0, 1\}$. If*

$$\sum_{k=1}^{m-1} d_k F_{k+1} = F_m,$$

then

$$\sum_{k=1}^{m-1} d_k F_k = F_{m-1}.$$

If instead

$$\sum_{k=1}^{m-1} d_k F_{k+1} = -F_m,$$

then

$$\sum_{k=1}^{m-1} d_k F_k = -F_{m-1}.$$

Proof. It is enough to prove the positive case, since the negative case follows by replacing each d_k with $-d_k$.

Let

$$\varphi := \frac{1+\sqrt{5}}{2}, \qquad \psi := \frac{1-\sqrt{5}}{2}.$$

By Binet's formula,

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}} \qquad (n \geq 1).$$

Define

$$P(z) := \sum_{k=1}^{m-1} d_k z^k.$$

The hypothesis reads

$$\frac{\varphi P(\varphi) - \psi P(\psi)}{\sqrt{5}} = F_m = \frac{\varphi^m - \psi^m}{\sqrt{5}},$$

so

$$C := \varphi P(\varphi) - \varphi^m = \psi P(\psi) - \psi^m.$$

The first expression is an algebraic integer in $\mathbb{Z}[\varphi]$, while the second is its Galois conjugate. Since they are equal, C is fixed by conjugation and therefore lies in $\mathbb{Q}$; hence C is an ordinary integer.

On the other hand,

$$|C| \leq \sum_{k=1}^{m-1} |\psi|^{k+1} + |\psi|^m < \sum_{j=2}^{\infty} |\psi|^j = 1.$$

Thus the integer C satisfies $|C| < 1$, so $C = 0$. Therefore

$$\varphi P(\varphi) = \varphi^m, \qquad \psi P(\psi) = \psi^m,$$

and hence

$$P(\varphi) = \varphi^{m-1}, \qquad P(\psi) = \psi^{m-1}.$$

Applying Binet's formula again gives

$$\sum_{k=1}^{m-1} d_k F_k = \frac{P(\varphi) - P(\psi)}{\sqrt{5}} = \frac{\varphi^{m-1} - \psi^{m-1}}{\sqrt{5}} = F_{m-1}.$$

□

Proposition 7.2 (Pairwise overlap reduction)**.** *For every $m \geq 3$ there is a well-defined map*

$$\Theta_m : \mathcal{L}_2(Y_m) \to X_{m-1}$$

with the following property. If

$$a = \mathrm{Fold}_m(pr), \qquad b = \mathrm{Fold}_m(rq)$$

for some bits $p, q \in \{0, 1\}$ and some overlap word $r = r_1 \cdots r_{m-1} \in \Omega_{m-1}$, then

$$\Theta_m(ab) = \mathrm{Fold}_{m-1}(r).$$

Proof. Take two raw words

$$prq, \qquad p'r'q'$$

of length $m + 1$ such that

$$\mathrm{Fold}_m(pr) = \mathrm{Fold}_m(p'r'), \qquad \mathrm{Fold}_m(rq) = \mathrm{Fold}_m(r'q').$$

Write

$$\Delta := N(r) - N(r').$$

Since $r, r' \in \Omega_{m-1}$, one has

$$|\Delta| \leq \sum_{k=1}^{m-1} F_{k+1} = F_{m+2} - 2 < F_{m+2}.$$

By Corollary 2.6, equality of the second folded symbols implies

$$\Delta + (q - q')F_{m+1} \in \{-F_{m+2}, 0, F_{m+2}\}.$$

Because $q - q' \in \{-1, 0, 1\}$ and $F_{m+2} = F_{m+1} + F_m$, this forces

$$\Delta \in \{0, \pm F_m, \pm F_{m+1}\}.$$

We claim that the cases $\Delta = \pm F_m$ are impossible. Suppose first that

$$\Delta = F_m.$$

Set

$$d_k := r_k - r'_k \in \{-1, 0, 1\} \qquad (1 \leq k \leq m - 1).$$

Then

$$\sum_{k=1}^{m-1} d_k F_{k+1} = F_m,$$

so Lemma 7.1 gives

$$\sum_{k=1}^{m-1} d_k F_k = F_{m-1}.$$

Hence

$$\begin{aligned} N(pr) - N(p'r') &= (p - p') + \sum_{k=1}^{m-1} d_k F_{k+2} \\ &= (p - p') + \sum_{k=1}^{m-1} d_k (F_{k+1} + F_k) \\ &= (p - p') + F_m + F_{m-1} \\ &= (p - p') + F_{m+1}. \end{aligned}$$

This lies in

$$\{F_{m+1} - 1, F_{m+1}, F_{m+1} + 1\},$$

which is disjoint from $\{-F_{m+2}, 0, F_{m+2}\}$ because $F_{m+2} = F_{m+1} + F_m > F_{m+1} + 1$. By Corollary 2.6, the first folded symbols cannot be equal, a contradiction. The case $\Delta = -F_m$ is identical after changing signs.

Therefore

$$\Delta \in \{0, \pm F_{m+1}\}.$$

By Corollary 2.6 at length $m - 1$, this is exactly the condition

$$\mathrm{Fold}_{m-1}(r) = \mathrm{Fold}_{m-1}(r').$$

Thus the folded overlap depends only on the pair of output symbols, and the map Θ_m is well defined. □

Remark 7.3. The restriction $m \geq 3$ is essential. For $m = 2$, both raw words 00 and 11 fold to 00, so the pair $(00, 00) \in \mathcal{L}_2(Y_2)$ does not determine the overlapping bit.

Definition 7.4 (Recursive overlap decoders)**.** For integers $m \geq 3$ and $1 \leq n \leq m$, define

$$\Theta_{m,n} : \mathcal{L}_n(Y_m) \to X_{m-n+1}$$

as follows. For $n = 1$, set

$$\Theta_{m,1}(a_1) := a_1.$$

For $(m, n) = (3, 2)$, set

$$\Theta_{3,2}(a_1 a_2) := \Theta_3(a_1 a_2).$$

For $(m, n) = (3, 3)$, let $\omega_1 \cdots \omega_5$ be the unique raw lift from Proposition 6.14 and define

$$\Theta_{3,3}(a_1 a_2 a_3) := \omega_3.$$

For $m \geq 4$ and $2 \leq n \leq m$, define

$$\Theta_{m,n}(a_1 \cdots a_n) := \Theta_{m-1,n-1}\big(\Theta_m(a_1 a_2)\, \Theta_m(a_2 a_3) \cdots \Theta_m(a_{n-1} a_n)\big).$$

Proposition 7.5 (Common-overlap decoding)**.** *Let $m \geq 3$ and $1 \leq n \leq m$. Suppose*

$$\omega = \omega_1 \cdots \omega_{m+n-1} \in \Omega_{m+n-1}, \qquad a_j := \mathrm{Fold}_m(\omega_j \cdots \omega_{j+m-1}) \quad (1 \leq j \leq n).$$

Then

$$\Theta_{m,n}(a_1 \cdots a_n) = \mathrm{Fold}_{m-n+1}(\omega_n \cdots \omega_m).$$

Thus $\Theta_{m,n}$ recovers the fold of the common overlap of the n consecutive raw windows.

Proof. We argue by induction on m.

If $m = 3$, there are three cases. For $n = 1$, the claim is the definition of $\Theta_{3,1}$. For $n = 2$, it is exactly Proposition 7.2. For $n = 3$, Proposition 6.14 gives a unique raw lift

$$\omega_1 \cdots \omega_5$$

of $a_1a_2a_3$, and by definition

$$\Theta_{3,3}(a_1a_2a_3) = \omega_3.$$

Since $X_1 = \{0, 1\}$ and Fold_1 fixes admissible words by Proposition 3.6(ii), this is

$$\mathrm{Fold}_1(\omega_3).$$

Thus the statement holds for $m = 3$.

Now assume $m \geq 4$ and that the proposition is known at level $m - 1$ for every admissible block length. The case $n = 1$ is again the definition. Assume $n \geq 2$. By Proposition 7.2, for each $1 \leq j \leq n - 1$ one has

$$\Theta_m(a_ja_{j+1}) = \mathrm{Fold}_{m-1}(\omega_{j+1} \cdots \omega_{j+m-1}).$$

These are precisely the $(m - 1)$-folds of the $n - 1$ consecutive length-$(m - 1)$ windows cut from the raw block

$$\omega_2 \cdots \omega_{m+n-2}.$$

Applying the induction hypothesis at level $m - 1$ to the $n - 1$ folded length-$(m - 1)$ windows yields

$$\Theta_{m,n}(a_1 \cdots a_n) = \Theta_{m-1,n-1}\big(\Theta_m(a_1a_2) \cdots \Theta_m(a_{n-1}a_n)\big) = \mathrm{Fold}_{m-n+1}(\omega_n \cdots \omega_m),$$

as claimed. □

Theorem 7.6 (Unique lift of a length-m output block)**.** *Let $m \geq 3$. Every block*

$$a_1 \cdots a_m \in \mathcal{L}_m(Y_m)$$

has a unique raw lift

$$\omega_1 \cdots \omega_{2m-1} \in \Omega_{2m-1}$$

satisfying

$$\mathrm{Fold}_m(\omega_j \cdots \omega_{j+m-1}) = a_j \qquad (1 \leq j \leq m).$$

Proof. Existence is immediate from the definition of $\mathcal{L}_m(Y_m)$.

For uniqueness, suppose

$$\omega_1 \cdots \omega_{2m-1}$$

is any lift of the given output block. For $1 \leq k \leq m$ define

$$L_k := \Theta_{m,m-k+1}(a_1 \cdots a_{m-k+1}), \qquad R_k := \Theta_{m,m-k+1}(a_k \cdots a_m).$$

By Proposition 7.5,

$$L_k = \mathrm{Fold}_k(\omega_{m-k+1} \cdots \omega_m), \qquad R_k = \mathrm{Fold}_k(\omega_m \cdots \omega_{m+k-1}).$$

For $k = 1$ this gives

$$L_1 = R_1 = \omega_m \in \{0, 1\},$$

so the central bit is determined.

Now fix $2 \le k \le m$ and assume inductively that the central blocks

$$\omega_{m-k+2} \cdots \omega_m, \qquad \omega_m \cdots \omega_{m+k-2}$$

have already been determined. The word L_k is the fold of the raw k-block

$$\omega_{m-k+1}\omega_{m-k+2} \cdots \omega_m,$$

whose suffix of length $k-1$ is known. By Proposition 6.5, applied with k in place of m, the two possible prepended bits 0 and 1 produce distinct folds. Therefore L_k determines the new left bit ω_{m-k+1} uniquely.

Similarly, R_k is the fold of

$$\omega_m \cdots \omega_{m+k-2}\omega_{m+k-1},$$

whose prefix of length $k-1$ is known. By Proposition 6.4, again with k in place of m, the appended bit is uniquely determined by R_k. Hence ω_{m+k-1} is fixed as well.

Proceeding outward from $k = 1$ to $k = m$ reconstructs all bits

$$\omega_1, \ldots, \omega_{2m-1}$$

uniquely. □

Corollary 7.7 (Uniform synchronizing length)**.** *Let $m \ge 3$. Every block in $\mathcal{L}_m(Y_m)$ is synchronizing for the presentation G_m. Moreover the block*

$$e_{m,0}e_{m,1} \cdots e_{m,m-2}$$

of length $m-1$ is not synchronizing. Consequently the minimal synchronizing length of G_m is exactly m.

Proof. By Theorem 7.6, every block in $\mathcal{L}_m(Y_m)$ is carried by exactly one length-m path in G_m, hence is synchronizing.

For the second claim, Proposition 6.13 constructs two sequences U_m and V_m whose first $m-1$ output symbols are

$$e_{m,0}e_{m,1} \cdots e_{m,m-2}$$

and coincide there. The corresponding length-$(m-1)$ paths in G_m end at the suffix vertices

$$010^{m-3} \qquad \text{and} \qquad 10^{m-2},$$

which are distinct. Thus the block admits at least two lifts with different terminal vertices, so it is not synchronizing. The distinct mth output symbols constructed in Proposition 6.13 show in particular that these two lifts do not merge after one further step. □

Theorem 7.8 (Finite-block losslessness in Fibonacci numeration)**.** *Let $m \ge 3$ and $n \ge m$. Define*

$$B_{m,n} : \Omega_{n+m-1} \to \mathcal{L}_n(Y_m), \qquad B_{m,n}(\omega_1 \cdots \omega_{n+m-1}) := a_1 \cdots a_n,$$

where

$$a_j := \mathrm{Fold}_m(\omega_j \cdots \omega_{j+m-1}) \qquad (1 \le j \le n).$$

Then $B_{m,n}$ is a bijection.

(m,n)	direct enumeration of $\lvert\mathcal{L}_n(Y_m)\rvert$	2^{n+m-1}
$(3,3)$	32	32
$(3,4)$	64	64
$(3,5)$	128	128
$(4,4)$	128	128
$(4,5)$	256	256
$(4,6)$	512	512

Table 3: Small-scale verification of Corollary 7.9 obtained by brute-force enumeration of all raw blocks in Ω_{n+m-1} and folding each consecutive length-m window.

Proof. Surjectivity is immediate from the definition of $\mathcal{L}_n(Y_m)$.

For injectivity, fix a block

$$a_1 \cdots a_n \in \mathcal{L}_n(Y_m).$$

Its prefix

$$a_1 \cdots a_m \in \mathcal{L}_m(Y_m)$$

has a unique raw lift by Theorem 7.6, hence it is carried by a unique length-m path in G_m. Since G_m is right-resolving by Proposition 6.4, each subsequent symbol $a_{m+1}, \ldots, a_n$ determines the next edge uniquely. Thus the whole block $a_1 \cdots a_n$ is carried by a unique length-n path in G_m.

Now a length-n path in G_m is uniquely determined by its initial vertex $v = v_1 \cdots v_{m-1} \in V_m$ together with the appended bits $b_1, \ldots, b_n \in \{0,1\}$. Equivalently, it is uniquely determined by the raw word

$$v_1 \cdots v_{m-1} b_1 \cdots b_n \in \Omega_{n+m-1}.$$

By construction of the labels in G_m, this raw word is precisely the unique length-$(n+m-1)$ block whose consecutive folded windows are $a_1, \ldots, a_n$. Therefore $B_{m,n}$ is injective. □

Corollary 7.9 (Exact block counts above the threshold)**.** *For every $m \geq 3$ and $n \geq m$,*

$$|\mathcal{L}_n(Y_m)| = 2^{n+m-1}.$$

In particular,

$$|\mathcal{L}_m(Y_m)| = 2^{2m-1}.$$

Proof. Theorem 7.8 identifies $\mathcal{L}_n(Y_m)$ with $\Omega_{n+m-1} = \{0,1\}^{n+m-1}$, whose cardinality is 2^{n+m-1}. □

Corollary 7.10 (Higher-block identification with the full two-shift)**.** *Let $m \geq 3$ and $n \geq m$. The bijection $B_{m,n}$ identifies the n-block presentation of Y_m with the $(n+m-1)$-block presentation of the full binary shift. Equivalently, consecutive blocks in $\mathcal{L}_n(Y_m)$ correspond exactly to consecutive overlapping blocks in Ω_{n+m-1}.*

Proof. Let

$$a = a_1 \cdots a_n, \qquad a' = a_2 \cdots a_{n+1}$$

be consecutive n-blocks in some point of Y_m. If

$$B_{m,n}^{-1}(a) = \omega_1 \cdots \omega_{n+m-1}, \qquad B_{m,n}^{-1}(a') = \omega'_1 \cdots \omega'_{n+m-1},$$

then both raw words are lifts of the same length-$(n+1)$ output block $a_1 \cdots a_{n+1} \in \mathcal{L}_{n+1}(Y_m)$. By Theorem 7.8, that length-$(n+1)$ output block has a unique raw lift in Ω_{n+m}, so necessarily

$$\omega'_1 \cdots \omega'_{n+m-2} = \omega_2 \cdots \omega_{n+m-1}.$$

Thus adjacency of n-blocks in Y_m is exactly overlap by $n+m-2$ bits of their raw lifts in Ω_{n+m-1}, which is the adjacency rule in the $(n+m-1)$-block presentation of the full binary shift. Hence the two higher-block presentations are canonically identified. □

Corollary 7.11 (Linear-time finite-block decoding)**.** *Fix $m \geq 3$. For every $n \geq m$, a block*

$$a_1 \cdots a_n \in \mathcal{L}_n(Y_m)$$

can be decoded to its unique raw lift in deterministic time $O_m(n)$ and working space $O(m)$ beyond the space used to store the output.

Proof. By Theorem 7.6, the initial block $a_1 \cdots a_m$ determines the unique raw lift

$$\omega_1 \cdots \omega_{2m-1}.$$

This fixes the current suffix vertex

$$\omega_{m+1} \cdots \omega_{2m-1} \in V_m.$$

For each $j = m+1, \ldots, n$, the current suffix vertex together with the next symbol a_j determines a unique outgoing edge of G_m by Proposition 6.4; following that edge reveals the next appended bit ω_{j+m-1}. Repeating this step decodes the entire raw lift. Since m is fixed, the initialization cost is $O_m(1)$, each new output symbol is processed in constant time, and only the current length-$(m-1)$ suffix together with the already decoded raw word must be stored. □

Remark 7.12 (Small decoding examples)*.* For $m = 3$, the output block

$$(a_1, a_2, a_3) = (001, 010, 100)$$

has unique raw lift 00100. Thus the central-bit inverse returns

$$\kappa_{3,1}(001, 010, 100) = 1.$$

For $m = 4$, the output block

$$(a_1, a_2, a_3, a_4) = (1001, 0010, 1000, 0000)$$

has unique raw lift 1110000. Hence the position-parameterized inverse family reads off

$$\kappa_{4,0} = 1, \qquad \kappa_{4,1} = 1, \qquad \kappa_{4,2} = 1, \qquad \kappa_{4,3} = 0$$

on this single normalized block. The first example is already fully admissible, while the second passes through nontrivial local normalization before decoding.

Theorem 7.13 (Exact finite-type order over the folded alphabet)**.** *For every $m \geq 3$, the shift Y_m is an m-step shift of finite type over the alphabet X_m, and it is not an $(m-1)$-step shift of finite type.*

Proof. Let

$$y = (y_t)_{t\in\mathbb{Z}} \in X_m^{\mathbb{Z}}$$

be such that every length-$(m+1)$ block of y lies in $\mathcal{L}_{m+1}(Y_m)$. For each $t \in \mathbb{Z}$, the length-m block

$$b_t := y_{t-m+1} \cdots y_t$$

belongs to $\mathcal{L}_m(Y_m)$ and is therefore synchronizing by Corollary 7.7. Let v_t denote the unique terminal vertex of a path in G_m labeled by b_t.

Now fix t. Since

$$y_{t-m+1} \cdots y_t y_{t+1} \in \mathcal{L}_{m+1}(Y_m),$$

there exists a path in G_m carrying this length-$(m+1)$ block. Its prefix of length m is labeled by b_t, so by uniqueness its terminal vertex is v_t. Its suffix of length m is labeled by

$$b_{t+1} = y_{t-m+2} \cdots y_{t+1},$$

so by the same uniqueness statement its terminal vertex is v_{t+1}. Hence the last edge of that path is an edge

$$v_t \xrightarrow{y_{t+1}} v_{t+1}.$$

Thus the vertices $(v_t)_{t\in\mathbb{Z}}$ form a bi-infinite path in G_m labeled by y, and therefore $y \in Y_m$. This proves that Y_m is m-step.

To show that Y_m is not $(m-1)$-step, let

$$b := e_{m,0} e_{m,1} \cdots e_{m,m-2} \in \mathcal{L}_{m-1}(Y_m).$$

By Corollary 7.7, the block b is not synchronizing, so reading b in G_m can end at at least two distinct vertices. By Proposition 6.13, the length-m block

$$bc = b\,010^{m-2}$$

belongs to $\mathcal{L}_m(Y_m)$. By Theorem 7.6, this length-m block is carried by exactly one path in G_m. Consequently, among the terminal vertices reached by reading b, exactly one admits an outgoing edge labeled 010^{m-2}. Since b is not synchronizing, at least one other terminal vertex reached by b does not admit that label.

Every finite path in G_m occurs in some bi-infinite labeled path because the underlying graph is the binary de Bruijn graph on words of length $m-1$ and is therefore strongly connected. Hence there exist two occurrences of the same length-$(m-1)$ block b in points of Y_m such that one occurrence may be followed by 010^{m-2} and another may not. Thus the admissible next symbol after b depends on more than the last $m-1$ output symbols, so Y_m is not an $(m-1)$-step shift of finite type. □

Theorem 7.14 (Local inverse rules)**.** *For every $m \geq 3$ and every $0 \leq r \leq m-1$, let*

$$\kappa_{m,r} : \mathcal{L}_m(Y_m) \to \{0,1\}$$

send a block $a_1 \cdots a_m$ to the $(r+1)$st bit of its unique raw lift from Theorem 7.6. Define

$$\Psi_m^{(r)} : Y_m \to \{0,1\}^{\mathbb{Z}}, \qquad (\Psi_m^{(r)}(y))_t := \kappa_{m,r}(y_{t-r} \cdots y_{t-r+m-1}).$$

Then $\Psi_m^{(r)}$ is a sliding block code with memory r and anticipation $m-1-r$, and

$$\Psi_m^{(r)} \circ \Phi_m = \mathrm{id}_{\{0,1\}^{\mathbb{Z}}}, \qquad \Phi_m \circ \Psi_m^{(r)} = \mathrm{id}_{Y_m}.$$

Equivalently, each length-m observation window yields an exact inverse rule for $\Phi_m : \{0,1\}^{\mathbb{Z}} \to Y_m$, and Φ_m is a topological conjugacy between the full binary shift and the image shift Y_m.

Proof. By definition, $(\Psi_m^{(r)}(y))_t$ depends only on the block $y_{t-r}\cdots y_{t-r+m-1}$, so $\Psi_m^{(r)}$ has memory r and anticipation $m-1-r$.

Let $s \in \{0,1\}^{\mathbb{Z}}$ and set $y := \Phi_m(s)$. For each $t \in \mathbb{Z}$, applying Theorem 7.6 to the output block

$$y_{t-r}\cdots y_{t-r+m-1}$$

shows that its unique raw lift is

$$s_{t-r}\cdots s_{t-r+2m-2}.$$

Therefore the $(r+1)$st bit of that lift is s_t, so

$$(\Psi_m^{(r)}(y))_t = s_t.$$

Hence

$$\Psi_m^{(r)}(\Phi_m(s)) = s,$$

so $\Psi_m^{(r)} \circ \Phi_m = \mathrm{id}$.

Now let $y \in Y_m$. By definition of Y_m there exists $s \in \{0,1\}^{\mathbb{Z}}$ with $\Phi_m(s) = y$. Applying the first identity to this s gives

$$\Psi_m^{(r)}(y) = \Psi_m^{(r)}(\Phi_m(s)) = s,$$

and therefore

$$\Phi_m(\Psi_m^{(r)}(y)) = \Phi_m(s) = y.$$

Thus Φ_m and $\Psi_m^{(r)}$ are mutual inverses. □

Corollary 7.15 (The one-sided memory threshold is exact)**.** *Let $m \geq 3$. Among all sliding block codes*

$$\Psi : Y_m \to \{0,1\}^{\mathbb{Z}}$$

with anticipation 0 *and*

$$\Psi \circ \Phi_m = \mathrm{id}_{\{0,1\}^{\mathbb{Z}}},$$

the smallest possible memory is exactly $m-1$.

Proof. The endpoint inverse $\Psi_m^{(m-1)}$ from Theorem 7.14 has memory $m-1$, while Proposition 6.13 shows that no one-sided left inverse can use smaller memory. □

7.2 Bernoulli measures and block statistics

We now pass from exact block transport to explicit probability laws. The block bijection $B_{m,n}$ identifies each admissible folded block with one raw binary block of length $n+m-1$, so Bernoulli product measures push forward to closed cylinder laws on the folded alphabet.

Theorem 7.16 (Bernoulli transport and exact cylinder law)**.** *Fix $m \geq 3$ and let ν_p be the Bernoulli product measure on $\{0,1\}^{\mathbb{Z}}$ with parameter $p \in (0,1)$. Set*

$$\mu_{m,p} := (\Phi_m)_*\nu_p.$$

For a finite word w in either shift space, write $[w]_0$ for the cylinder on the coordinates $0,\ldots,|w|-1$. Then for every $n \geq m$ and every $a \in \mathcal{L}_n(Y_m)$,

$$\mu_{m,p}([a]_0) = \nu_p([\omega(a)]_0) = p^{|\omega(a)|_1}(1-p)^{n+m-1-|\omega(a)|_1}, \qquad \omega(a) := B_{m,n}^{-1}(a),$$

where $|\omega(a)|_1$ denotes the number of 1's in the raw block $\omega(a)$.

In particular, when $p = \frac{1}{2}$ one has

$$\mu_{m,1/2}([a]_0) = 2^{-(n+m-1)} \qquad (a \in \mathcal{L}_n(Y_m),\ n \geq m),$$

so every admissible folded block of length $n \geq m$ is equally likely.

Proof. By definition of the pushforward measure,

$$\mu_{m,p}([a]_0) = \nu_p(\Phi_m^{-1}([a]_0)).$$

Now $\Phi_m(s) \in [a]_0$ means precisely that the raw block $s_0 \cdots s_{n+m-2} \in \Omega_{n+m-1}$ maps to a under $B_{m,n}$. Since Theorem 7.8 makes $B_{m,n}$ bijective, this is equivalent to

$$s_0 \cdots s_{n+m-2} = \omega(a) := B_{m,n}^{-1}(a).$$

Hence

$$\Phi_m^{-1}([a]_0) = [\omega(a)]_0.$$

The Bernoulli product law of that raw cylinder is

$$\nu_p([\omega(a)]_0) = p^{|\omega(a)|_1}(1-p)^{n+m-1-|\omega(a)|_1},$$

which proves the formula. The case $p = \frac{1}{2}$ is immediate. □

Corollary 7.17 (Exact entropy profile and excess entropy)**.** *Let $(Y_t)_{t\in\mathbb{Z}}$ be the coordinate process on Y_m under $\mu_{m,p}$, and write*

$$h_2(p) := -p\log_2 p - (1-p)\log_2(1-p).$$

Then for every $n \geq m$,

$$H_{\mu_{m,p}}(Y_0^{n-1}) = (n+m-1)h_2(p).$$

Consequently

$$h_{\mu_{m,p}}(\sigma) = h_2(p), \qquad E(\mu_{m,p}) = (m-1)h_2(p),$$

where $E(\mu_{m,p})$ denotes the excess entropy. In particular, for $p = \frac{1}{2}$ one has

$$H_{\mu_{m,1/2}}(Y_0^{n-1}) = n+m-1, \qquad E(\mu_{m,1/2}) = m-1.$$

Proof. Fix $n \geq m$. By Theorem 7.16, the map $a \mapsto \omega(a) = B_{m,n}^{-1}(a)$ is a bijection from $\mathcal{L}_n(Y_m)$ onto Ω_{n+m-1}, and the probability of the cylinder $[a]_0$ under $\mu_{m,p}$ is exactly the Bernoulli probability of the raw cylinder $[\omega(a)]_0$. Therefore

$$H_{\mu_{m,p}}(Y_0^{n-1}) = -\sum_{a\in\mathcal{L}_n(Y_m)} \mu_{m,p}([a]_0)\log_2 \mu_{m,p}([a]_0)$$

equals the Shannon entropy of a Bernoulli block of length $n+m-1$, namely $(n+m-1)h_2(p)$.

Dividing by n and letting $n \to \infty$ gives $h_{\mu_{m,p}}(\sigma) = h_2(p)$. The exact linear formula also yields

$$E(\mu_{m,p}) = \lim_{n\to\infty}\big(H_{\mu_{m,p}}(Y_0^{n-1}) - nh_{\mu_{m,p}}(\sigma)\big) = (m-1)h_2(p).$$

The case $p = \frac{1}{2}$ follows from $h_2(\frac{1}{2}) = 1$. □

Theorem 7.18 (Exact Markov order under Bernoulli transport)**.** *Fix $m \geq 3$ and $p \in (0,1)$. For every block*

$$\beta = a_1 \cdots a_m \in \mathcal{L}_m(Y_m),$$

let $v(\beta) \in V_m$ denote the terminal suffix state of the unique path in G_m carrying β. Then for every $t \in \mathbb{Z}$ and every $b \in \{0,1\}$,

$$\mu_{m,p}\big(Y_{t+1} = \mathrm{Fold}_m(v(\beta)b) \,\big|\, Y_{t-m+1} \cdots Y_t = \beta\big) = p^b(1-p)^{1-b}.$$

In particular, under $\mu_{m,p}$ the coordinate process $(Y_t)_{t\in\mathbb{Z}}$ is an m-step Markov process. It is not an $(m-1)$-step Markov process.

Proof. By shift invariance it suffices to prove the conditional formula at time $t = m-1$. Fix

$$\beta = a_1 \cdots a_m \in \mathcal{L}_m(Y_m),$$

and let

$$\omega(\beta) = \omega_1 \cdots \omega_{2m-1} := B_{m,m}^{-1}(\beta).$$

The terminal suffix state of the unique path carrying β is therefore

$$v(\beta) = \omega_{m+1} \cdots \omega_{2m-1} \in V_m.$$

By Theorem 7.16, the event $\{Y_0 \cdots Y_{m-1} = \beta\}$ pulls back under Φ_m to the raw cylinder

$$\{S_0 \cdots S_{2m-2} = \omega(\beta)\},$$

where $(S_t)_{t\in\mathbb{Z}}$ is the coordinate process on the Bernoulli shift. Conditioned on that raw cylinder, the bits $S_m, \ldots, S_{2m-2}$ are fixed and equal to $v(\beta)$, while the next raw bit S_{2m-1} remains independent of the past with

$$\nu_p(S_{2m-1} = b) = p^b(1-p)^{1-b}.$$

The next folded symbol is exactly

$$Y_m = \mathrm{Fold}_m(v(\beta)S_{2m-1}),$$

so

$$\mu_{m,p}\big(Y_m = \mathrm{Fold}_m(v(\beta)b) \,\big|\, Y_0 \cdots Y_{m-1} = \beta\big) = p^b(1-p)^{1-b}.$$

This proves the displayed formula. Since Proposition 6.4 shows that the two labels $\mathrm{Fold}_m(v(\beta)0)$ and $\mathrm{Fold}_m(v(\beta)1)$ are distinct, the law of the next symbol is determined entirely by the last m output symbols. Hence (Y_t) is m-step Markov under $\mu_{m,p}$.

To see that the Markov order is not smaller, suppose instead that (Y_t) were an $(m-1)$-step Markov process. Theorem 7.16 shows that every admissible cylinder in Y_m has positive $\mu_{m,p}$-measure, so $\mu_{m,p}$ has full support on Y_m. For a stationary process with full support, an $(m-1)$-step Markov law has support equal to the $(m-1)$-step shift of finite type obtained by declaring a length-m block admissible exactly when it has positive measure: by repeated conditioning, a longer block has positive probability if and only if each of its consecutive length-m subblocks does. Because every admissible length-m block of Y_m has positive measure, that support would be Y_m itself. This contradicts Theorem 7.13, which states that Y_m is not an $(m-1)$-step shift of finite type. Therefore the Markov order is exactly m. □

Corollary 7.19 (Conjugacy invariants)**.** *For every $m, n \geq 3$, the shifts Y_m and Y_n are topologically conjugate. In particular, for every $m \geq 3$,*

$$h_{\mathrm{top}}(Y_m) = \log 2, \qquad \#\mathrm{Fix}(\sigma^n|_{Y_m}) = 2^n \quad (n \geq 1), \qquad \zeta_{Y_m}(z) = \frac{1}{1-2z}.$$

Proof. By Theorem 7.14, every Y_m with $m \geq 3$ is conjugate to the full two-shift. The conjugacy

$$Y_m \xrightarrow{\Psi_m^{(m-1)}} \{0,1\}^{\mathbb{Z}} \xrightarrow{\Phi_n} Y_n$$

gives the first claim, and the entropy, fixed-point counts, and zeta function are the corresponding invariants of the full two-shift. □

7.3 The natural right-resolving presentation

We now return to the subset construction. The exact decoding theorem shows that the determinized presentation has a finite transient ambiguity shell and that its essential core is already visible in the explicit graph G_m. The next results quantify the resulting deterministic presentation size on the natural folded alphabet X_m.

Definition 7.20 (Deterministic candidate-set presentation)**.** For a nonempty subset $S \subset V_m$ and a label $a \in X_m$, define

$$\delta_m(S, a) := \{v' \in V_m : \exists\, e : v \to v' \text{ with } v \in S \text{ and } \ell_m(e) = a\}.$$

Starting from the state V_m, take all reachable nonempty subsets of V_m and add labeled edges

$$S \xrightarrow{a} \delta_m(S, a)$$

whenever $\delta_m(S, a) \neq \varnothing$. The resulting finite labeled graph is denoted H_m^{det}.

Proposition 7.21 (Deterministic right-resolving presentation)**.** *The graph H_m^{det} is right-resolving and presents the same labeled language as G_m. Consequently its edge shift has label image Y_m.*

Proof. By construction, for a fixed state S and label a, there is at most one outgoing edge with label a, so H_m^{det} is right-resolving. The subset construction preserves the accepted labeled language of a finite automaton, hence H_m^{det} and G_m present the same set of finite label words. Their bi-infinite label shifts therefore coincide, and by Theorem 6.2 this common image is Y_m. □

Proposition 7.22 (The singleton core)**.** *Let $m \geq 3$. Then:*

(i) *every state of H_m^{det} reached after at least m input symbols is a singleton;*

(ii) *the reachable singleton states are exactly the sets $\{v\}$ with $v \in V_m$;*

(iii) *the labeled subgraph induced by the singleton states is canonically labeled-isomorphic to G_m.*

Proof. Take any block

$$a_1 \cdots a_m \in \mathcal{L}_m(Y_m).$$

By Theorem 7.6, this block has a unique raw lift, hence a unique length-m path in G_m carrying that label word. Its terminal vertex is therefore uniquely determined by the block. Consequently, every reachable state after reading m symbols is a singleton. Since singletons are forward closed

under the subset construction, every state reached after at least m symbols is again a singleton. This proves (i).

For (ii), it remains to show that every singleton $\{v\}$ with $v \in V_m$ is reachable. By Theorem 6.11, the symbol b_m sends the full candidate set to a singleton. The underlying directed graph of G_m is the binary de Bruijn graph on words of length $m-1$, so it is strongly connected. Thus for every $v \in V_m$ there exists a path in G_m from that singleton state to v, and reading its label from the singleton yields $\{v\}$.

Finally, (iii) is immediate from the definitions: for singleton states,

$$\delta_m(\{v\}, a) = \{v'\}$$

if and only if G_m has an edge $v \to v'$ labeled a. Hence the singleton subgraph is exactly a relabeling of G_m. □

Proposition 7.23 (The ambiguity shell is transient)**.** *Let $m \geq 3$. Every non-singleton state of $H_m^{\det}$ is transient. Equivalently, the subgraph induced by the non-singleton states is acyclic.*

Proof. Suppose there were a directed cycle consisting entirely of non-singleton states. Since every state of $H_m^{\det}$ is reachable from the initial state V_m, one could first reach that cycle and then traverse it often enough to obtain a path of length at least m ending at a non-singleton state. This contradicts Proposition 7.22(i). Hence no such cycle exists, and every non-singleton state is transient. □

Proposition 7.24 (Exact nilpotency of the ambiguity shell)**.** *Let $m \geq 3$. Order the states of $H_m^{\det}$ by placing the non-singleton states first and the singleton core last. Then the adjacency matrix of $H_m^{\det}$ has the form*

$$A(H_m^{\det}) = \begin{pmatrix} N_m & R_m \\ 0 & A(G_m) \end{pmatrix}$$

for some matrix R_m, and

$$N_m^m = 0, \qquad N_m^{m-1} \neq 0.$$

Consequently

$$\det(I - zA(H_m^{\det})) = \det(I - zA(G_m)).$$

Proof. By Proposition 7.22(iii), the singleton states induce a labeled copy of G_m. Since singleton states are forward closed, there are no edges from the singleton core back into the non-singleton shell, so the adjacency matrix has the displayed block upper triangular form.

By Proposition 7.22(i), every path of length m in $H_m^{\det}$ ends in a singleton. Hence there is no path of length m contained entirely in the non-singleton shell, which is equivalent to $N_m^m = 0$.

By Corollary 7.7, there exists a non-synchronizing block of length $m-1$. Reading that block from the initial state V_m yields a path of length $m-1$ ending at a non-singleton state. Because singleton states are forward closed, every intermediate state on that path is also non-singleton. Therefore the shell contains a path of length $m-1$, so $N_m^{m-1} \neq 0$.

Finally,

$$\det(I - zA(H_m^{\det})) = \det(I - zN_m)\det(I - zA(G_m)).$$

Since N_m is nilpotent, $\det(I - zN_m) = 1$, and the determinant identity follows. □

Proposition 7.25 (The explicit graph is follower separated)**.** *Let $m \geq 3$. Distinct vertices of G_m have distinct follower sets.*

Proof. Let

$$b_m = u_1 \cdots u_m$$

be the alternating symbol from Definition 6.8. For each vertex

$$v = v_1 \cdots v_{m-1} \in V_m,$$

consider the length-m path obtained by appending the fixed bit word $u_1 \cdots u_m$. Let

$$\Lambda(v) \in \mathcal{L}_m(Y_m)$$

denote its label word.

We claim that $\Lambda(v)$ is carried by exactly one path in G_m, namely the one that starts at v and appends $u_1, \ldots, u_m$. Indeed, the last symbol of $\Lambda(v)$ is b_m, and by Theorem 6.11, the symbol b_m is carried by a unique edge. Thus the final edge of any path carrying $\Lambda(v)$ is fixed. Since G_m is one-step left-resolving by Proposition 6.5, the penultimate label then determines the penultimate edge uniquely; iterating this backward reconstruction shows that the whole path carrying $\Lambda(v)$ is unique.

In particular, its initial vertex is uniquely determined by $\Lambda(v)$. Hence if $v \neq w$, the word $\Lambda(v)$ belongs to the follower set of v but not to that of w. Therefore distinct vertices have distinct follower sets. □

Corollary 7.26 (The graph G_m is the canonical minimal right-resolving presentation)**.** *For every $m \geq 3$, the canonical minimal right-resolving presentation of Y_m is labeled-isomorphic to G_m. Equivalently, G_m is the right Fischer cover of Y_m.*

Proof. By Theorem 7.14, the shift Y_m is irreducible (indeed conjugate to the full two-shift). Proposition 7.21 gives a deterministic right-resolving presentation of Y_m. By Proposition 7.23, the only essential component of that presentation is the singleton core from Proposition 7.22, which is labeled-isomorphic to G_m. Proposition 7.25 shows that this core is already follower separated. The standard construction of the canonical minimal right-resolving presentation, equivalently of the right Fischer cover, as the follower-separated essential component of a deterministic right-resolving presentation therefore leaves G_m unchanged; see [**?**, Chs. 3–4] and [**?**, Ch. 4]. □

Theorem 7.27 (Size of the right-resolving presentation)**.** *For every $m \geq 3$, the canonical minimal right-resolving presentation of Y_m over the folded alphabet X_m has exactly*

$$|V(G_m)| = 2^{m-1}, \qquad |E(G_m)| = 2^m$$

states and edges. In particular, no right-resolving presentation of Y_m over X_m uses fewer than 2^{m-1} states. Since

$$|X_m| = F_{m+2},$$

one has

$$\frac{|V(G_m)|}{|X_m|} = \frac{2^{m-1}}{F_{m+2}} \sim \frac{\sqrt{5}}{2\varphi^2}\Big(\frac{2}{\varphi}\Big)^m, \qquad \varphi = \frac{1+\sqrt{5}}{2}.$$

Thus, although every Y_m with $m \geq 3$ is conjugate to the full two-shift, its canonical deterministic presentation over the natural folded alphabet has exponentially growing state complexity, even relative to the alphabet size.

Proof. Corollary 7.26 identifies G_m as the canonical minimal right-resolving presentation of Y_m. The state and edge counts are therefore those of G_m, and Definition 6.1 gives

$$|V(G_m)| = 2^{m-1}, \qquad |E(G_m)| = 2^m.$$

Minimality of the Fischer cover yields the state lower bound for all right-resolving presentations over X_m.

The alphabet size is

$$|X_m| = F_{m+2}$$

by Corollary 2.2. Using the Binet asymptotic $F_{m+2} \sim \varphi^{m+2}/\sqrt{5}$ gives

$$\frac{2^{m-1}}{F_{m+2}} \sim \frac{2^{m-1}\sqrt{5}}{\varphi^{m+2}} = \frac{\sqrt{5}}{2\varphi^2}\Big(\frac{2}{\varphi}\Big)^m,$$

which proves the claimed representation gap. □

Corollary 7.28 (Coincidence of decoding and presentation scales)**.** *For every $m \geq 3$, the following quantities are exact and all equal to the same scale:*

(i) *the finite-type order of Y_m over the alphabet X_m is m;*

(ii) *the minimal synchronizing length of the presentation G_m is m;*

(iii) *the nilpotency index of the ambiguity-shell matrix N_m is m;*

(iv) *the minimal memory of a one-sided sliding-block inverse of Φ_m is $m-1$.*

In particular, although the shifts Y_m with $m \geq 3$ are all conjugate to the full two-shift, their natural presentations over the folded alphabets X_m still reflect the window length. Theorem 7.18 shows that the same threshold is also the Markov memory under every transported Bernoulli law, while Theorem 7.27 quantifies the accompanying exponential presentation gap on the natural alphabet.

Proof. Part (i) is Theorem 7.13, part (ii) is Corollary 7.7, part (iii) is Proposition 7.24, and part (iv) is Corollary 7.15. □

8 Arithmetic reformulations and consequences

The results of the preceding sections admit purely number-theoretic formulations that do not refer to shift spaces, factor maps, or finite-state presentations. Throughout this section fix $m \geq 3$, $n \geq m$, and write $L := n + m - 1$.

8.1 Local Fibonacci congruence rigidity

For a binary block $\omega = \omega_1 \cdots \omega_L \in \{0,1\}^L$ and $1 \leq t \leq n$, define the t-th local Fibonacci weighted sum

$$S_t^{(m)}(\omega) := \sum_{j=0}^{m-1} \omega_{t+j} F_{j+2},$$

and its standard residue modulo F_{m+2},

$$R_t^{(m)}(\omega) \in \{0, 1, \ldots, F_{m+2} - 1\}, \qquad R_t^{(m)}(\omega) \equiv S_t^{(m)}(\omega) \pmod{F_{m+2}}.$$

Collect all local residues into a vector

$$\mathcal{R}_{m,n}(\omega) := \big(R_1^{(m)}(\omega), \dots, R_n^{(m)}(\omega)\big) \in \{0, 1, \dots, F_{m+2} - 1\}^n.$$

The key observation is that by Corollary 2.6,

$$R_t^{(m)}(\omega) = V_m\big(\mathrm{Fold}_m(\omega_t \cdots \omega_{t+m-1})\big),$$

where $V_m : X_m \to \{0, 1, \dots, F_{m+2} - 1\}$ is the bijection from Proposition 2.5.

Theorem 8.1 (Local Fibonacci congruence rigidity)**.** *The map $\mathcal{R}_{m,n} : \{0,1\}^L \to \{0, 1, \dots, F_{m+2} - 1\}^n$ is injective. Equivalently, for any $(r_1, \dots, r_n) \in \{0, 1, \dots, F_{m+2} - 1\}^n$, the congruence system*

$$\sum_{j=0}^{m-1} \omega_{t+j} F_{j+2} \equiv r_t \pmod{F_{m+2}} \qquad (1 \le t \le n)$$

has at most one solution $\omega \in \{0,1\}^L$. The image $\mathcal{R}_{m,n}(\{0,1\}^L)$ has exactly $2^L = 2^{n+m-1}$ elements.

Proof. Take any $\omega \in \{0,1\}^L$. For each t, set $a_t := \mathrm{Fold}_m(\omega_t \cdots \omega_{t+m-1}) \in X_m$. By Corollary 2.6, $V_m(a_t) \equiv S_t^{(m)}(\omega) \pmod{F_{m+2}}$, and since $V_m(a_t)$ already lies in $\{0, 1, \dots, F_{m+2} - 1\}$, one has $V_m(a_t) = R_t^{(m)}(\omega)$. Thus

$$\mathcal{R}_{m,n}(\omega) = \big(V_m(a_1), \dots, V_m(a_n)\big).$$

Suppose $\mathcal{R}_{m,n}(\omega) = \mathcal{R}_{m,n}(\omega')$. Then $V_m(a_t) = V_m(a'_t)$ for each t. Since V_m is a bijection (Proposition 2.5), $a_t = a'_t$ for every $1 \le t \le n$. Hence ω and ω' produce the same length-n folded block under $B_{m,n}$. By Theorem 7.8, $B_{m,n}$ is a bijection, so $\omega = \omega'$. The image has 2^L elements because $\mathcal{R}_{m,n}$ is an injection from a set of size 2^L. □

8.2 Fingerprint separation of equi-valued representations

Corollary 8.2 (Fingerprint separation)**.** *For each integer N, let*

$$\Omega_L(N) := \Big\{\omega \in \{0,1\}^L : \sum_{k=1}^{L} \omega_k F_{k+1} = N\Big\}.$$

Then $\mathcal{R}_{m,n}$ restricted to $\Omega_L(N)$ is still injective:

$$|\Omega_L(N)| = \big|\{\mathcal{R}_{m,n}(\omega) : \omega \in \Omega_L(N)\}\big|.$$

Proof. Since $\mathcal{R}_{m,n}$ is injective on all of $\{0,1\}^L$ (Theorem 8.1), it is injective on every subset. □

In other words, all length-L binary Fibonacci representations of a single integer N have mutually distinct overlapping local Zeckendorf fingerprints.

8.3 Exact bit recovery from residue windows

Write $\mathfrak{R}_{m,\ell} := \mathcal{R}_{m,\ell}(\{0,1\}^{\ell+m-1})$ for the set of realizable local residue vectors of length ℓ.

Theorem 8.3 (Bit recovery from local congruence data)**.** *For each $0 \le s \le m-1$, there exists a function $\rho_{m,s} : \mathfrak{R}_{m,m} \to \{0,1\}$ such that for every $\omega \in \{0,1\}^{n+m-1}$ and every $1 \le t \le n-m+1$,*

$$\omega_{t+s} = \rho_{m,s}\big(R_t^{(m)}(\omega), R_{t+1}^{(m)}(\omega), \dots, R_{t+m-1}^{(m)}(\omega)\big).$$

In particular, taking $s = m-1$ gives a one-sided decoding rule $g_m := \rho_{m,m-1}$ with

$$\omega_t = g_m\big(R_{t-m+1}^{(m)}(\omega), \dots, R_t^{(m)}(\omega)\big).$$

No rule using fewer than m consecutive residues can recover all input bits.

Proof. For any admissible m-tuple $(r_1, \dots, r_m) \in \mathfrak{R}_{m,m}$, set $a_i := V_m^{-1}(r_i) \in X_m$ and define

$$\rho_{m,s}(r_1, \dots, r_m) := \kappa_{m,s}(a_1 \cdots a_m),$$

where $\kappa_{m,s}$ is the local inverse rule from Theorem 7.14.

Now take any $\omega \in \{0,1\}^{n+m-1}$ and fix t. Set $a_j := \mathrm{Fold}_m(\omega_j \cdots \omega_{j+m-1})$ for $t \le j \le t+m-1$. Then $R_j^{(m)}(\omega) = V_m(a_j)$, so $a_j = V_m^{-1}(R_j^{(m)}(\omega))$. By Theorem 7.14, the folded block $a_t \cdots a_{t+m-1}$ has unique raw lift $\omega_t \cdots \omega_{t+2m-2}$, and $\kappa_{m,s}$ reads its $(s+1)$-st bit. Therefore

$$\rho_{m,s}\big(R_t^{(m)}(\omega), \dots, R_{t+m-1}^{(m)}(\omega)\big) = \kappa_{m,s}(a_t \cdots a_{t+m-1}) = \omega_{t+s}.$$

For the optimality claim, suppose a rule using only $k+1 < m$ consecutive residues existed. Since V_m is a symbol-by-symbol bijection, this would yield a sliding-block left inverse $\Psi : Y_m \to \{0,1\}^{\mathbb{Z}}$ with memory $k < m-1$, contradicting Corollary 7.15. □

8.4 Distribution of random local residue vectors

Theorem 8.4 (Exact distribution of random local residues)**.** *Let $X_1, \dots, X_L$ be independent with $\mathbb{P}(X_i = 1) = p$ and $\mathbb{P}(X_i = 0) = 1-p$, where $0 < p < 1$. Define the random local residue vector $\mathbf{R}_{m,n} := \mathcal{R}_{m,n}(X_1 \cdots X_L)$. Then for every $\mathbf{r} = (r_1, \dots, r_n) \in \{0, 1, \dots, F_{m+2}-1\}^n$,*

$$\mathbb{P}(\mathbf{R}_{m,n} = \mathbf{r}) = \begin{cases} p^{|\omega(\mathbf{r})|_1}(1-p)^{L-|\omega(\mathbf{r})|_1}, & \mathbf{r} \in \mathfrak{R}_{m,n}, \\ 0, & \mathbf{r} \notin \mathfrak{R}_{m,n}, \end{cases}$$

where $\omega(\mathbf{r})$ is the unique preimage guaranteed by Theorem 8.1. When $p = \frac{1}{2}$, $\mathbf{R}_{m,n}$ is uniformly distributed on $\mathfrak{R}_{m,n}$:

$$\mathbb{P}(\mathbf{R}_{m,n} = \mathbf{r}) = 2^{-L} \qquad (\mathbf{r} \in \mathfrak{R}_{m,n}).$$

Proof. If $\mathbf{r} \notin \mathfrak{R}_{m,n}$, then no $\omega \in \{0,1\}^L$ satisfies $\mathcal{R}_{m,n}(\omega) = \mathbf{r}$, so the probability is 0.

Suppose $\mathbf{r} \in \mathfrak{R}_{m,n}$. Set $a_t := V_m^{-1}(r_t) \in X_m$ for $1 \le t \le n$. The event $\{\mathbf{R}_{m,n} = \mathbf{r}\}$ is equivalent to the folded block being $a_1 \cdots a_n$. By Theorem 7.8, this folded block has a unique raw lift $\omega(\mathbf{r}) \in \{0,1\}^L$. Hence

$$\{\mathbf{R}_{m,n} = \mathbf{r}\} = \{(X_1, \dots, X_L) = \omega(\mathbf{r})\}.$$

Since $X_1, \dots, X_L$ are independent Bernoulli(p),

$$\mathbb{P}(\mathbf{R}_{m,n} = \mathbf{r}) = p^{|\omega(\mathbf{r})|_1}(1-p)^{L-|\omega(\mathbf{r})|_1}.$$

When $p = \frac{1}{2}$, every $\omega \in \{0,1\}^L$ has probability 2^{-L}. □

Corollary 8.5 (Entropy of random local residues)**.** *Under the assumptions of Theorem* 8.4*, for every* $n \geq m$*,*

$$H(\mathbf{R}_{m,n}) = (n + m - 1)h_2(p),$$

where $h_2(p) := -p \log_2 p - (1 - p) \log_2(1 - p)$*. In particular, when* $p = \frac{1}{2}$*,* $H(\mathbf{R}_{m,n}) = n + m - 1$*.*

Proof. By Theorem 8.4, the distribution of $\mathbf{R}_{m,n}$ is the image of a length-L Bernoulli block under the bijection $\mathcal{R}_{m,n}$. Entropy is invariant under bijective relabeling, so $H(\mathbf{R}_{m,n}) = H(X_1, \ldots, X_L) = Lh_2(p) = (n + m - 1)h_2(p)$. □

8.5 Markov order of the residue process

Theorem 8.6 (Exact Markov order of the local residue process)**.** *Let* $(X_t)_{t\in\mathbb{Z}}$ *be a two-sided Bernoulli*(p) *process with* $0 < p < 1$*. Define*

$$R_t := \left(\sum_{j=0}^{m-1} X_{t+j} F_{j+2} \right) \bmod F_{m+2}, \qquad R_t \in \{0, 1, \ldots, F_{m+2} - 1\}.$$

Then $(R_t)_{t\in\mathbb{Z}}$ *is an exact* m*-step Markov process: given any realizable block* $(r_{t-m+1}, \ldots, r_t)$*, the conditional law of* R_{t+1} *depends only on these* m *values, and no* $(m - 1)$*-step Markov law can describe* (R_t)*.*

More precisely, let $a_i := V_m^{-1}(r_{t-m+i}) \in X_m$*,* $\beta := a_1 \cdots a_m \in \mathcal{L}_m(Y_m)$*, and let* $v(\beta) \in \{0, 1\}^{m-1}$ *be the terminal suffix of the unique raw lift of* β*. Then for each* $b \in \{0, 1\}$*,*

$$\mathbb{P}(R_{t+1} = V_m(\mathrm{Fold}_m(v(\beta)b)) \mid R_{t-m+1} = r_{t-m+1}, \ldots, R_t = r_t) = p^b(1 - p)^{1-b}.$$

Proof. Set $Y_t := \mathrm{Fold}_m(X_t \cdots X_{t+m-1}) \in X_m$. By Corollary 2.6 and Proposition 2.5, $R_t = V_m(Y_t)$ for all t. Since V_m is a bijection, (R_t) and (Y_t) are related by a symbol-by-symbol relabeling.

Theorem 7.18 states that (Y_t) is an exact m-step Markov process under $\mu_{m,p}$, with the displayed transition probabilities. Translating through V_m yields the same conclusion for (R_t): the event $\{Y_j = a_{j-(t-m)}\}$ is equivalent to $\{R_j = r_j\}$ because $R_j = V_m(Y_j)$ and V_m is invertible. Likewise $Y_{t+1} = \mathrm{Fold}_m(v(\beta)b)$ if and only if $R_{t+1} = V_m(\mathrm{Fold}_m(v(\beta)b))$.

If (R_t) were $(m - 1)$-step Markov, then $(Y_t) = V_m^{-1}(R_t)$ would also be $(m - 1)$-step Markov, contradicting Theorem 7.18. □

Competing interests

The author(s) declare none.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used OpenAI ChatGPT and Anthropic Claude in order to assist with language polishing and editing of the manuscript text. After using these tools, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.